Calibrated Hallucination Risk Estimation Under Domain and Task Shift Using Evidence-Consistent XGBoost

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Abstract

Large language models answer fluently, but some answers lack evidence support. Detectors rely on a large judge, an encoder, or model internals. These options are costly and rarely report calibration under shift. Feature-based detectors are cheaper, but their calibration and length sensitivity are seldom measured together, and HaluEval length is almost a label proxy. This study tests whether a simple feature model can estimate risk and how its calibration and ranking change under shift. EC-XGB combines 26 base features, eight claim-level NLI aggregates, and a source indicator. It is trained on HaluEval QA and RAGTruth rows with grouped splits and three seeds, and evaluated with a fixed-hyperparameter ablation, source and target Platt calibration, conformal prediction, SHAP stability, a length-only baseline, an external ensemble, and judge stress tests. The fixed-hyperparameter EC-XGB reaches F1 0.9854 and AUROC 0.9986 on HaluEval QA. Lexical overlap and length carry the in-domain score: removing them costs 0.0171 and 0.0066 F1. On MedHallu, the model ranks the paired hallucinated answer above the grounded one in 86.7% of 1,000 pairs (AUROC 0.821), but flags every row and is badly calibrated (ECE 0.493). On HaluEval dialogue and summarization, it fails in different ways: dialogue is flagged almost entirely, and summarization ranks below a length-only model. On RAGTruth QA it reaches AUROC 0.558. Target calibration lowers ECE from 0.731 to 0.125, but the calibrated score flags no target rows at a 0.5 threshold. Split conformal reaches 0.944 coverage at a 90% target, but positive coverage is only 0.685, and FaithBench coverage falls to 0.320. The GPT-5.6 Luna judge reaches F1 0.565 on natural QA, where EC-XGB reaches 0.310. On a 158-row length-matched set EC-XGB reaches 0.907 and the judge 0.865, but the interval includes zero. EC-XGB is a domain-specific risk score, not a general-purpose detector.

1. Introduction

Black-box detectors score a candidate answer from its text and context. SelfCheckGPT samples several answers and checks whether they agree [1]. Luna replaces the large judge with a smaller evaluation model [2]. Semantic entropy clusters sampled answers by meaning [3]. A hybrid framework combines frozen encoder features with lightweight classifiers [4]. Judges carry high computational costs, and encoder models are hard to interpret. Two recent reviews group detectors into feature-based, encoder-based, and hybrid approaches [5, 6]. Both name evaluation and reliability as open problems, and a third review reaches the same conclusion about reliability [7].

HaluAgent showed that small models can select tools, but it needs multiple calls per answer [8]. Valentin et al. proposed a calibrated multi-score detector with a cost analysis, but it relies on LLM scoring [9]. Sundaragiri et al. combined retrieval, NLI, and XGBoost, but report only classification metrics [10]. Haq et al. assigned SHAP-LIME attributions to each word, but did not measure their reliability [11]. Deng et al. found that detectors degrade outside their training domain, but focus on multi-turn dialogue [12]. FaithBench shows that modern LLMs still produce summarization hallucinations, but does not measure detector calibration [13].

The existing literature leaves four issues unaddressed. Calibration is infrequently reported for lightweight detectors, explanation quality is rarely measured, cross-domain behavior is rarely tested, and error patterns are rarely analyzed case by case. The conventional feature-based baseline also has limits. HaluEval is saturated and the score depends on answer length. Its calibration also breaks under shift. This paper addresses these gaps with a feature-based model that scores the output of a black-box LLM. It takes a query q, a reference context c, and a candidate answer a, and outputs $P(\text{hallucinated} \mid q, c, a)$. EC-XGB combines 26 base features in seven groups, eight claim-level NLI aggregates, and a source indicator in a tuned XGBoost classifier. Platt calibration maps the score to a probability, SHAP explains

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feature contributions, and grouped splits with length and overlap controls reduce shortcut solutions. Calibration is fitted on validation rows only, and the target-calibration experiments measure the reliability problem. The claim-level aggregates support per-claim explanations, and the measured transfer gain comes from multi-source training rather than from those aggregates. Transfer limits are established on RAGTruth and FaithBench. The evaluation adds MedHallu [14], a HaluBench subset [15] with no training overlap, the HaluEval task shifts, conformal prediction, and judge stress tests. The added evidence separates ranking from calibration: the model transfers its ranking to medical pairs but not its probability, and it fails under a task shift.

HaluEval QA provides 10,000 question pairs, each with a correct and a hallucinated answer, for training and in-domain testing. RAGTruth provides natural responses for adaptation and recalibration. The target use case is question answering with a reference context, and FaithBench serves only as an out-of-domain summarization stress test. The fixed-hyperparameter EC-XGB (m3) reaches F1 0.9854 and AUROC 0.9986 on HaluEval QA. RAGTruth shows the same pattern: the flag rate falls on the tasks seen in training, while held-out QA stays near chance. On MedHallu the same model ranks the paired hallucinated answer above the grounded one in 86.7% of 1,000 pairs but flags every row. On HaluEval dialogue it flags almost every row, and on HaluEval summarization it ranks below a length-only model. Target calibration on 5,034 RAGTruth QA calibration rows lowers ECE from 0.731 to 0.125, but it does not change the ranking. One answer takes 127.5 ms on GPU (2.00 s on CPU) at about $0.001 per 1,000 predictions.

2. Conventional Method

2.1. Preliminaries

This section describes the conventional feature-based detector and the criteria used to evaluate it. A hallucination is an answer that the reference context does not support, or that the context contradicts. Each row is a triple (q_i, c_i, a_i) of question, reference context, and candidate answer, with label $y_i \in \{0, 1\}$. A detector is usually scored with precision, recall, F1, or the area under the receiver operating characteristic curve (AUROC) on labeled rows. A deployed detector faces two more demands. Calibration requires the predicted risk to match the observed frequency, and expected calibration error (ECE) summarizes the gap across confidence bins [16]. Explanation requires the warning to name the features that contributed to it, and Shapley additive explanations (SHAP) provide that for a tabular model [17]. This work measures both properties on the black-box model.

2.2. Baseline Method

The conventional method is a tuned XGBoost classifier [18] over the 26 base features in Table 1, trained on HaluEval QA. XGBoost models nonlinear interactions and gives exact TreeSHAP values, which the explanation analysis needs. A published feature-based detector uses the same learner with retrieval and NLI features [10], so this model is the natural baseline. Trees are added one at a time, and each tree fits the current gradient and curvature of the loss. A row enters at the root of each tree, follows the split conditions to one leaf per tree, and the leaf weights are summed into a raw margin. A logistic link maps the raw margin to a probability.

Let $x_i \in \mathbb{R}^{26}$ be the feature vector of row i, $y_i \in \{0, 1\}$ its label, and n the number of training rows. The model sums K trees into a raw margin, $z_i = \sum_{k=1}^{K} f_k(x_i)$. The training objective combines a differentiable loss with a regularizer over the trees,

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l(y_i, z_i) + \sum_{k=1}^{K} \left(\gamma T_k + \tfrac{1}{2}\lambda \|w_k\|^2\right), \tag{1}$$

where $\phi = \{f_k\}_{k=1}^{K}$ collects the trees, f_k is the k-th tree with T_k leaves and leaf-weight vector $w_k \in \mathbb{R}^{T_k}$, γ is the penalty per leaf, and λ is an L2 weight penalty. The loss $l(y_i, z_i)$ is the logistic loss on the raw margin. The probability is the sigmoid of the raw margin, $\hat{p}_i = \sigma(z_i) = 1/(1 + e^{-z_i})$.

At boosting step t, a new tree f_t with T leaves and leaf weights $w_1, \ldots, w_T$ is added, and the objective is approximated, up to a constant, by a second-order expansion around the current raw margin,

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \left[g_i f_t(x_i) + \tfrac{1}{2} h_i f_t^2(x_i)\right] + \gamma T + \tfrac{1}{2}\lambda \sum_{j=1}^{T} w_j^2, \tag{2}$$

where $g_i = \partial l(y_i, z)/\partial z$ and $h_i = \partial^2 l(y_i, z)/\partial z^2$ are evaluated at $z_i^{(t-1)}$, and $f_t(x_i) = w_{q(x_i)}$ is the leaf weight that row i reaches, with $q(x_i)$ the leaf index. For the logistic loss, $g_i = \hat{p}_i - y_i$ and $h_i = \hat{p}_i(1 - \hat{p}_i)$.

For a fixed tree structure, the optimal weight of leaf j is

$$w_j^* = -\frac{G_j}{H_j + \lambda}, \tag{3}$$

where $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} h_i$ are the sums over the rows I_j that reach leaf j. The gain of a split is

$$\text{Gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda}\right] - \gamma, \tag{4}$$

with G_L and H_L for the left child and G_R and H_R for the right child. A split is kept only when the gain is positive. The tree also stops at the maximum depth or when a leaf has fewer than the minimum child weight.

Depth, learning rate, tree count, and the row and column subsampling rates are tuned on the 14,000 training rows. Each seed runs its own 30-configuration search with stratified group five-fold cross-validation, and the grouping keeps both answers to the same question in one fold. The seed-42 values are in Section 4, and the per-seed values are in the repository. Each seed trains on the 14,000 rows with early stopping on the 3,000 validation rows (30 rounds). The 300-tree setting is the tuned upper bound, and seed 42 stops at 209 trees. The headline comparison uses the uncalibrated probability at a fixed 0.5 threshold on the 3,000 test rows. Platt and isotonic maps are fitted on the same validation rows for the calibration study. The source-only P0 threshold is selected on those rows at a 5% false-positive budget. The test rows are never used for tuning, early stopping, calibration, or threshold selection. The final step returns the mean over the three seeds. The search leaves γ, λ, and the minimum child weight at their XGBoost defaults.

Algorithm 1 Baseline feature-based XGBoost (training and evaluation)

Require: labeled rows (q_i, c_i, a_i, y_i) and seeds $S = \{42, 123, 456\}$
Ensure: trained model, test metrics, and Platt and isotonic maps
1: extract the 26 base features for every row
2: **for** each seed $s \in S$ **do**
3: tune XGBoost on the 14,000 training rows with 30 random configurations and stratified group 5-fold cross-validation
4: train the classifier on the same rows with early stopping on the 3,000 validation rows
5: evaluate the raw score on the 3,000 test rows at the fixed 0.5 threshold
6: fit Platt and isotonic maps on the validation rows for the calibration study
7: **end for**
8: **return** the mean metrics over the seeds

2.3. Limitations and Research Gap

The baseline has three method limits. The score depends on answer length, it has no claim-level evidence, and it uses a source-fitted calibrator. It also trains on HaluEval alone. Two evaluation limits sit beside them: HaluEval is saturated, and the source probability does not transfer to a new corpus [19]. Section 3 addresses the method limits with log-scaled lengths and monotone constraints, eight claim-level aggregates, RAGTruth non-QA rows, and a source indicator. The calibration protocol addresses the shift limit, and the evaluation reports ranking, calibration, and paired errors rather than F1 alone. The log transform compresses the length scale but does not remove the cue. The gap is a risk score that keeps its ranking and probability when the corpus changes.

3. Proposed Method

3.1. Working Principle

Figure 1 shows the pipeline. The system takes a question, a reference context, and a candidate answer, and extracts 35 features: 26 base features in seven groups, eight claim-level NLI aggregates, and a source indicator. EC-XGB maps the feature vector to a raw score, and a Platt calibrator fitted on the RAGTruth QA calibration rows maps the score to the displayed probability. The system reports the risk score, per-claim verdicts, and SHAP contributions. The lower band shows the deployment path: a Next.js front end calls a FastAPI service that loads the model, calibrator, explainer, and feature models at startup. The rightmost block lists the evaluation corpora. HaluEval measures in-domain performance,

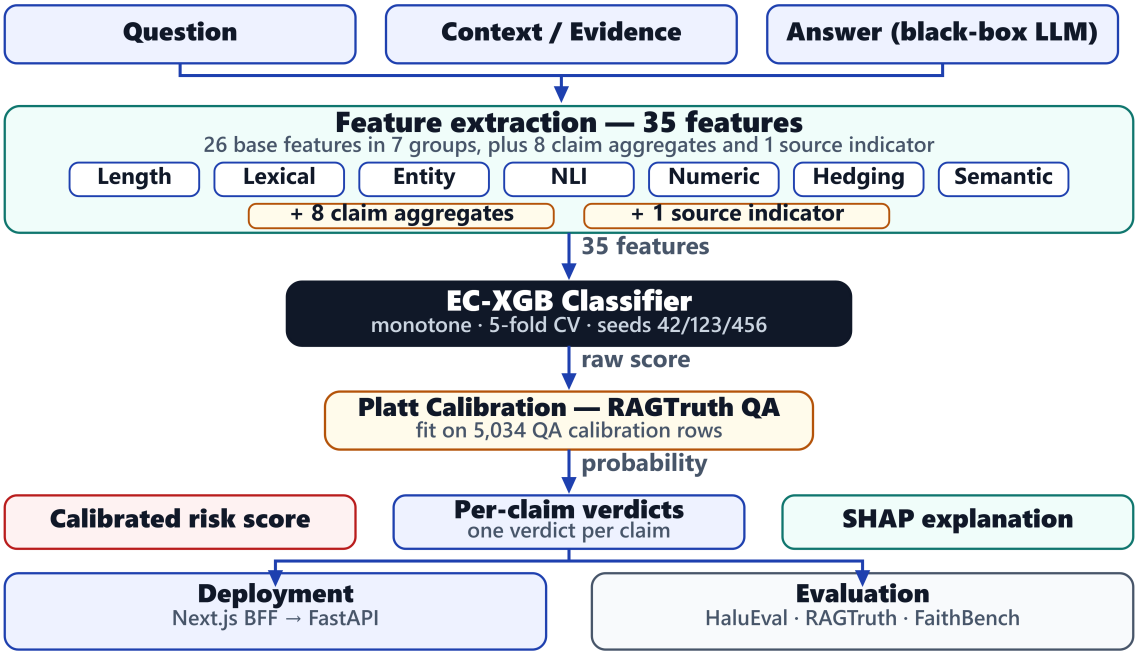


Figure 1: End-to-end pipeline of EC-XGB.

RAGTruth summarization and data-to-text measure adaptation to task types seen in training, and RAGTruth QA and FaithBench are held out.

The task is to estimate $P(\text{hallucinated} \mid q, c, a)$ for a question q, reference context c (if available), and a candidate answer a from a black-box LLM. The target is not truthfulness in general but evidence consistency: whether a follows from c or contradicts it. The output is a probability, and a decision threshold turns it into a flag.

3.2. Features

The model uses 26 base features in seven groups (Table 1) and eight claim-level aggregates (Table 2). NLI probabilities come from a DeBERTa-v3 cross-encoder [20] trained on SNLI [21] and MultiNLI [22], semantic similarity uses MiniLM embeddings [23, 24], and entities come from spaCy [25]. Length is included because HaluEval labels longer answers as hallucinated more often, so it is a strong but fragile signal. Lexical overlap captures copying and is the strongest SHAP feature, but it cannot detect a fluent paraphrase that contradicts the evidence. A novel entity is a useful warning sign, although a correct answer can introduce a new name. The NLI features are the only group that represents logical support rather than surface overlap. A changed number is a common and easy-to-detect hallucination. Hedging words are weak on their own but separate a cautious correct answer from a confident wrong one. Semantic similarity catches paraphrase and topic drift that lexical overlap misses. The eight claim aggregates repeat the NLI signals at the level of atomic claims, which is the level a reviewer checks. RAGChecker also uses entailment-based claim metrics [26].

3.3. Evidence-Consistent XGBoost

EC-XGB changes the conventional setup in three places. First, the character and word counts are log-scaled, and monotone constraints are placed on the evidence features. The constraints are negative on context overlap, context Jaccard, context entailment, context cosine, claim minimum and mean entailment, and the supported-claim ratio, and positive on context contradiction, novel-entity ratio, novel numbers, claim maximum and mean contradiction, and the contradicted-claim ratio. All other features are unconstrained. Second, the eight claim features in Table 2 are added: seven NLI aggregates and a claim count. They use the same NLI scorer and support-first rule as the verification layer. A claim is supported when any retained sentence has an entailment probability of at least 0.5 and at least as high as its contradiction probability. It is contradicted when no sentence supports it and a sentence has a contradiction probability of at least 0.5 and above its entailment probability. Otherwise it is unsupported.

Table 1
Base feature groups (26 features total).

Group	Features	Source
Length/style	n_chars, n_words, n_sentences, avg_word_len	regex
Lexical	overlap vs. context & question, Jaccard ×2	token sets
Entity	entity counts, overlap ratio, novel-entity ratio	spaCy NER
NLI	entail/contradict/neutral × 2 directions	cross-encoder NLI
Numeric	number counts, overlap ratio, novel numbers	regex
Hedging	hedge count, hedge density	lexicon
Semantic	cosine(context,answer), cosine(question,answer)	MiniLM

Table 2
Claim-level NLI aggregates (8 features).

Signal	Features	Source
Contradiction	max and mean contradiction, contradicted-claim ratio	NLI cross-encoder
Support	supported-claim ratio, unsupported-claim ratio, min and mean entailment	NLI cross-encoder
Decomposition	number of claims	sentence splitter

Note. The answer is split into at most six atomic clauses. Conjunctions such as "and" or "but" are broken into separate clauses. Each clause is scored against its four best-matching context sentences, and the pair probabilities are aggregated.

Consider an answer with K claims. Let $C(k)$ be the maximum contradiction probability of claim k over its selected context sentences, $E(k)$ the maximum entailment probability, and v_k its verdict. The aggregate vector is

$$\psi(x) = [\max_k C(k), \overline{C}, r_{\text{con}}, r_{\text{sup}}, r_{\text{unsup}}, \min_k E(k), \overline{E}, K], \tag{5}$$

with

$$r_{\text{con}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{contradicted}], \quad r_{\text{sup}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{supported}], \quad r_{\text{unsup}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{unsupported}]. \tag{6}$$

Here $x = (q, c, a)$ is the input triplet, and $\overline{C}$ and $\overline{E}$ are the means of $C(k)$ and $E(k)$ over the claims. For $K > 0$, the three ratios sum to one. The ablation in Section 5 shows that the block adds no aggregate F1 gain. The model input has 35 dimensions: the 26 base features (two of them log-scaled), the eight claim features, and one corpus indicator. The claim count K is kept because an answer with many claims is harder to verify.

Third, the training set adds the RAGTruth summarization and data-to-text rows. A binary corpus indicator is one for those rows and zero for HaluEval, so the classifier can separate the two sources. The classifier never sees RAGTruth QA rows, so its QA ranking is zero-shot. The display calibrator does see labeled QA rows below, so the display score is not zero-shot. The combination is called EC-XGB. Seed 42 is the deployed checkpoint, and the other seeds are a stability check.

Algorithm 2 lists the procedure. Steps 1 and 2 build the feature matrix. Steps 3 to 9 repeat the fit for the three seeds: each repetition builds the 35-feature input, trains on HaluEval plus the RAGTruth non-QA rows, fits the display calibrator on the 5,034 QA calibration rows, and selects the source and target thresholds. The QA rows stay out of classifier training, which keeps the QA ranking zero-shot.

Algorithm 2 Evidence-consistent XGBoost (EC-XGB) training and deployment

Require: HaluEval QA training and validation rows, RAGTruth summarization and data-to-text training rows, RAGTruth QA calibration rows, seeds $\mathcal{S} = \{42, 123, 456\}$
Ensure: seed-42 model and display calibrator, source and target thresholds, per-seed metrics
extract the 26 base features and the eight claim features for every row
log-scale the character and word counts, and attach the monotone constraints
for each seed $s \in \mathcal{S}$ **do**
    build the 35-feature input, with the corpus indicator at zero for HaluEval and one for RAGTruth
    train XGBoost on the HaluEval rows plus the RAGTruth non-QA rows, and keep every RAGTruth QA row out of classifier training
    fit the Platt display calibrator on the RAGTruth QA calibration rows
    select the source threshold on the HaluEval validation split
    select the target threshold on the QA calibration rows at a 5% false-positive budget
end for
return the seed-42 model and calibrator, both thresholds, and the per-seed metrics

3.4. Calibration and Explainability

Source calibration uses the HaluEval validation split. Platt scaling fits a sigmoid $p(x) = \sigma(Af(x) + B)$ to the uncalibrated probability $f(x)$, where A and B are fit by maximum likelihood [27]. Isotonic regression fits a non-decreasing step function g so that $p(x) = g(f(x))$ [28]. Both are compared with the uncalibrated probability on the HaluEval test split. The comparison reports expected calibration error over 10 equal-width bins,

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \tag{7}$$

where $\mathrm{acc}(B_m)$ and $\mathrm{conf}(B_m)$ are the empirical positive rate and mean confidence in bin B_m, the Brier score $\mathrm{BS} = \frac{1}{n}\sum_i (\hat{p}_i - y_i)^2$, and reliability diagrams [16]. Post-hoc calibration degrades when the target distribution moves [29].

The second protocol fits the calibrator on labeled target rows and evaluates it on a disjoint target test split. The deployed EC-XGB display score uses a Platt calibrator fitted on the 5,034 RAGTruth QA calibration rows. Those rows are scored with the corpus indicator set to one, and the API uses the same value for every request. This target calibration needs labeled target rows and is not zero-shot. The source-only protocol (P0 in Table 4) fits both the calibrator and the threshold on the HaluEval validation split. The deployed calibrator is Platt. An earlier deployment fit compared Platt and isotonic on the test split, and Section 5 records that choice.

A TreeExplainer computes exact SHAP values [17] for the raw XGBoost model. Global importance is the mean absolute SHAP value over a stratified test sample, and the explanation endpoint returns the five features with the largest absolute value for one prediction. Post-hoc attributions can be unstable even when the model is fixed [30], so the pilot evaluates them directly [31]. A 20-case pilot applies six operators: entity, number, and date replacement, support removal, context insertion, and clause shuffling. It reports the mean absolute score change, the fraction of large changes, the top-1 flip rate, and the Spearman rank correlation between the original and perturbed rankings.

3.5. System and Reproducibility

The method is deployed in a web application with a Next.js front end and a FastAPI backend. The API loads the EC-XGB model, the Platt calibrator, the SHAP explainer, and the NLI, spaCy, and MiniLM feature models at startup. The deployed checkpoint is the 35-feature m3 model with seed 42. Section 5 compares it with the conventional B2 checkpoint and with B6 m0, a 26-feature reference from the same tuning protocol that lacks the EC-XGB changes. Claims are split from the answer and checked with sentence-level NLI against the provided context, a local passage index, or a web search API. A claim is supported when any retained sentence has an entailment probability of at least 0.5 and at least as high as its contradiction probability. A contradiction decides only when no sentence supports the claim. A relevance gate drops sentences with no claim vocabulary, and the verdicts are shown per claim.

Figure 2 shows the deployment view. The browser calls the Next.js BFF, which streams chat tokens and forwards tool calls to the model. The analyze route calls FastAPI through the `/api/ml/*` rewrite. FastAPI loads EC-XGB, the Platt calibrator, SHAP, NLI, spaCy, and MiniLM at startup. The lower band lists the inference components. A tool

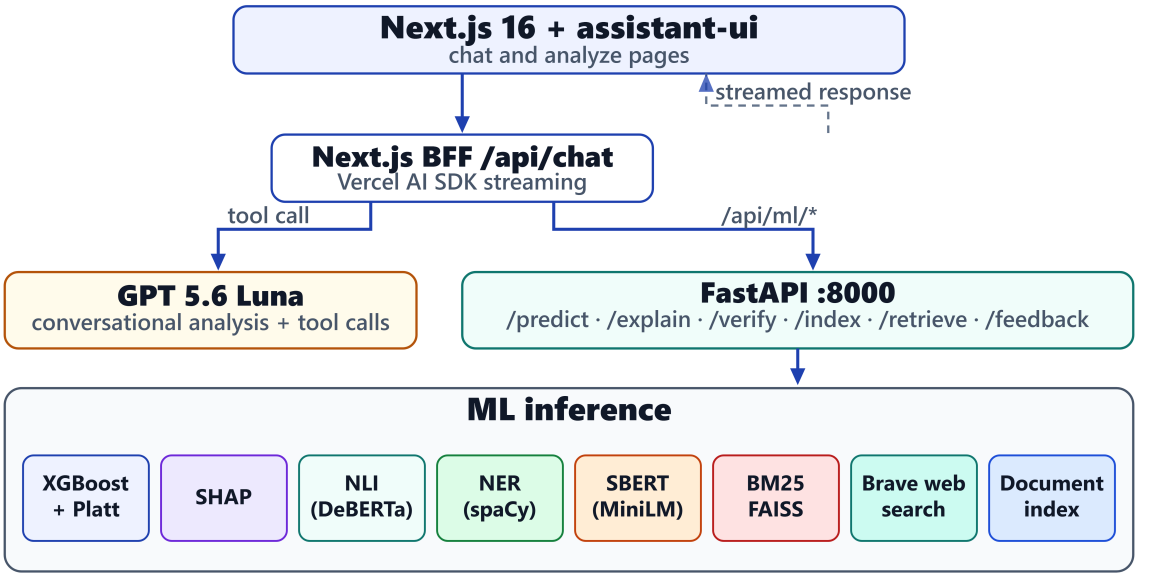


Figure 2: Deployment architecture of the HaluRISC system.

call goes to the judge, and a prediction request goes to `/predict`, `/explain`, or `/verify`. Figure 1 shows how one answer is scored, and Figure 2 shows where each component runs.

The card's headline follows the claim verdicts when they are present. A contradicted claim sets high risk, an unsupported claim sets medium risk, and a fully supported answer sets low risk. The percentage is the display score: the chat card adjusts it by the claim mix, and the analyze card reports the unadjusted Platt score.

Figure 3 shows the hallucinated-number case. The context says 5 days per decade and the answer says 10. The raw score is 0.999 and the target-calibrated score is 35.4%, which the card rounds to 35% and places in the medium band (0.35–0.60). The claim view marks the number claim unsupported, so the chat card also reports medium risk. The two labels agree here, but they come from different rules. The claim-verdict audit is reported in Section 5. The requirements file pins every dependency, split indices are stored with the artifacts, and the audit scripts check the rebuilt and unchanged reported values against the frozen artifacts. The audits run after every build and fail on any mismatch.

The contribution is the training and evaluation protocol, not a new learner. EC-XGB combines three standard changes: log-scaled length counts with monotone constraints, claim-level NLI aggregates for explanation, and multi-source training with a corpus indicator. Section 5 measures what each change contributes. The claim aggregates are kept for per-claim explanation and do not add an aggregate F1 gain. The seen-task gain comes from the m3 stack that adds the RAGTruth rows and the corpus indicator, and the two are not separated. The risk score needs no LLM judge calls. The optional claim judge is a separate verification step, and the reported EC-XGB results do not use it.

4. Experiments

4.1. Experimental Setup

All experiments were run on a Windows laptop with an NVIDIA RTX 3060 6 GB GPU, CUDA 12.8, torch 2.11.0+cu128, and 32 GB of RAM. The code runs on Python 3.12 with scikit-learn 1.9.0, XGBoost 3.3.0, SHAP 0.52.0, spaCy 3.8.14, and sentence-transformers 5.6.1. Each run can be reproduced because of saved split indices and centralized seeds. A randomized search was performed over 30 hyperparameter configurations with grouped five-fold cross-validation. For the seed-42 conventional model, the configuration has a maximum tree depth of 5, a learning rate of 0.05, 300 trees, a subsampling ratio of 0.7, and a column sampling ratio of 0.7. Each seed is tuned separately for the main comparison, and the component ablation in Section 5 holds one configuration fixed across variants so that the feature effects are not confounded with tuning. Seeds 42, 123, and 456 are repeated where a seed applies; the single-seed tables are labeled.

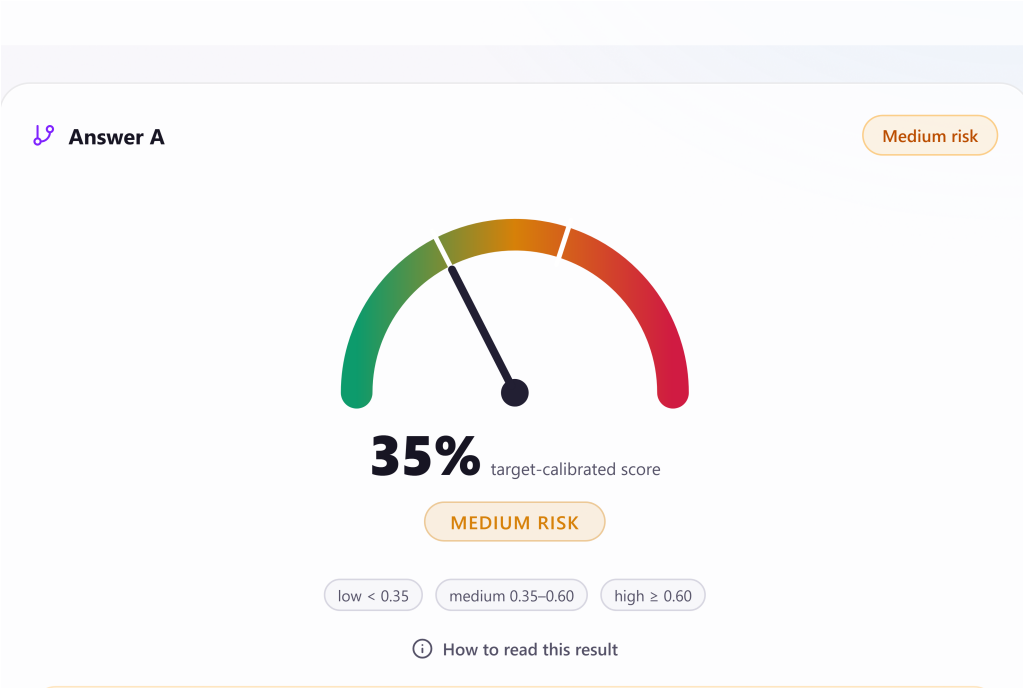


Figure 3: Risk card for the hallucinated-number case, showing the target-calibrated display score.

4.2. Dataset

HaluEval QA [32] provides 10,000 questions, each with a knowledge passage, a correct answer, and a hallucinated answer, so 20,000 labeled rows. The split is 70/15/15 at the item level, giving 14,000/3,000/3,000 rows, so both answers to the same question stay in one split. HaluEval is entirely ChatGPT-generated [32, 33], so the boundary between a factual error and a hallucination is not always clean. RAGTruth [34] supplies natural responses with human span annotations, and FaithBench [13] supplies modern-LLM summarization hallucinations. RAGTruth keeps its official document-level split: summarization has 4,758 training and 900 test rows, data-to-text has 5,298 and 900, and QA has 5,934 rows in 989 groups, with 900 test rows in 150 groups and 5,034 calibration rows. The three 900-row test splits are held out, giving a pooled 2,700-row test set. Table 3 lists the corpora and their positive rates.

Two further corpora and a task-shift pair test external validity without fitting or threshold selection. MedHallu pqa_labeled [14] gives 1,000 medical QA pairs with human-annotated gold answers and paired LLM-written hallucinations (2,000 rows, 50.0% positive). HaluBench [15] contributes a stratified 30% sample of its DROP, FinanceBench, and covidQA rows: 900 rows, 300 per source, balanced at 150 PASS and 150 FAIL. Its HaluEval, RAGTruth, and PubMedQA rows are excluded. The labels are HaluBench PASS/FAIL decisions, and the DROP and FinanceBench rows were converted to QA by the HaluBench authors. A seeded sample of 1,000 pairs from each HaluEval task gives 2,000 rows per task at 50.0% positive. This is a task-shift test, not a second independent QA corpus.

Leading and trailing whitespace is trimmed, and lexical matching is case-folded. NER and NLI read the original text, and entity strings are lowercased for matching. A missing context is a special case: lexical overlap is 0 and NLI probabilities are uniform. In m1 and later variants, the two length counts are log1p-transformed; m0 and the conventional model keep raw counts. Claims are split into at most six clause-level segments. When the context has more than four sentences, each claim is matched to its four best sentences by token overlap, and each side of a pair is truncated to 400 characters. HaluEval is split 70/15/15 by question item, RAGTruth keeps its official document-level split, and both index sets are frozen. A RAGTruth answer is positive when it carries at least one annotated hallucination span. FaithBench severity annotations are reduced by a worst-severity rule, and a Questionable annotation counts as positive. HaluEval uses its native label. NLI and SBERT inference ran on the reference GPU in fp16, while lexical and entity features run on the CPU.

Table 3
Datasets used for training and evaluation.

Corpus	Rows	Groups	Positive	Use
HaluEval QA	20,000	10,000	50.0%	train / validation / test
RAGTruth QA	5,934	989	17.8 / 31.1%	held-out test + calibration
RAGTruth full	17,790	2,965	34.9%	train + pooled test (2,700)
FaithBench	750	750	68.1%	zero-shot
MedHallu	2,000	1,000 pairs	50.0%	zero-shot external
HaluBench subset	900	3 sources	50.0%	zero-shot external
HaluEval dialogue	2,000	1,000 pairs	50.0%	task-shift test
HaluEval summarization	2,000	1,000 pairs	50.0%	task-shift test

Note. RAGTruth QA is a subset of RAGTruth full: its 900 test rows and 5,034 calibration rows are already counted there, so the rows are not additive. Its positive rate is 17.8% on the test split and 31.1% on the calibration split. The pooled RAGTruth test split holds the 900 summarization, 900 data-to-text, and 900 QA test rows. The remaining 15,090 rows form the official training split, and the 5,034 QA rows in it serve as the calibration block. The five zero-shot corpora (FaithBench, MedHallu, HaluBench, and the two task-shift sets) are never used for fitting or threshold selection.

The feature-based logistic baselines standardize their inputs on the training split; the TF-IDF models use sparse features, and XGBoost does not scale. The source indicator is 1 for FaithBench and MedHallu and 0 for HaluBench and the task-shift sets. FaithBench and RAGTruth summarization and data-to-text have no question, so question overlap is 0 and question similarity is the cosine between the answer and the encoded empty string. Licenses and the data-availability statement are in the Declarations.

4.3. Baselines and Experimental Procedure

Twelve models are tested. The heuristic sets risk to one minus lexical overlap, and a length-only logistic control uses the two log-scaled length counts. Logistic regression and a 300-tree random forest are the classical comparators. The conventional XGBoost model [18] uses the 26 base features. Three further controls isolate the feature design: NLI-only, answer-only TF-IDF, and full-input TF-IDF. A majority classifier, a fine-tuned RoBERTa-base [35], and a reduced reimplementation of the lightweight ensemble of Sundaragiri et al. [10] complete the set; the two judges are in Section 5. Internal-state detectors are outside the black-box setting [36]. The RoBERTa baseline receives up to 256 tokens and trains for three epochs.

EC-XGB's display calibrator is fitted on the RAGTruth QA split; the in-domain Platt calibrator is fitted on validation. Metrics are reported to four decimals and rates to one. Significance uses McNemar on seed-42 predictions, 2,000 two-level bootstrap resamples, and Wilcoxon across seeds. No question appears in both training and evaluation. Calibration is compared on raw, Platt, and isotonic scores; Table 4 defines P0, P1, and P2, and every protocol uses the same test rows.

Table 4
The three evaluation protocols.

Protocol	Calibrator fit	Threshold fit	Zero-shot
P0 source-only	HaluEval validation (3,000)	HaluEval validation	yes
P1 target-calibrated	RAGTruth QA calibration (5,034)	fixed at 0.5	no
P2 target-thresholded	RAGTruth QA calibration (5,034)	same rows at 5% FPR	no

Note. Only P0 is zero-shot on the target corpus. P1 and P2 read labels from the target calibration split. The 900-row test split is disjoint from that calibration split in every row.

The deployed system uses P1, and P2 is the thresholded alternative. SHAP [17] TreeExplainers give global and local explanations. Two checks test reliability: the rank agreement between the m3 and B2 SHAP importances and a six-operator perturbation pilot. Feature latency is measured on 200 HaluEval rows on the reference laptop. The 900 RAGTruth QA rows are scored by two judges, GPT-5.6 Luna and Qwen 3.5 4B, under the LLM-as-a-judge protocol [37], and costs are computed from the recorded token counts.

4.4. Evaluation Metrics

Two families of metrics are reported. Threshold-dependent metrics measure the decision at a fixed cut-off, and threshold-free metrics measure how well a model ranks answers. Both are needed because a model can rank well and still make poor decisions when the threshold is fixed. Precision is the share of flagged answers that are truly hallucinated. Recall is the share of hallucinated answers that the detector catches, and F1 is their harmonic mean. The false-positive rate is the share of correct answers that are flagged, which is the operating constraint under distribution shift. The flag rate is the share of answers above the decision threshold, and a rate near 100% means every answer reaches a human reviewer.

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad F_1 = \frac{2PR}{P+R}, \tag{8}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}}, \qquad \mathrm{Flag} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[\hat{p}_i \geq \tau]. \tag{9}$$

The second group measures ranking, calibration, and paired error patterns. AUROC is the probability that a randomly chosen hallucinated answer receives a higher score than a randomly chosen correct answer, with 0.5 at chance. PR-AUC is the area under the precision-recall curve, and it is more sensitive than AUROC when the positive class is rare. RAGTruth QA has a positive rate of 17.8%, so PR-AUC is reported beside AUROC. MCC uses all four cells of the confusion matrix and ranges from −1 to 1, with zero at chance. The Wilson interval gives a 95% range for a proportion and is used for the P2 recall and false-positive rate, where the counts are small.

$$\mathrm{AUROC} = \int_0^1 \mathrm{TPR}(u)\, d\,\mathrm{FPR}(u) = P(s^+ > s^-), \tag{10}$$

$$\text{PR-AUC} = \int_0^1 P(r)\, dr, \qquad \mathrm{MCC} = \frac{\mathrm{TP}\cdot\mathrm{TN} - \mathrm{FP}\cdot\mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}, \tag{11}$$

$$\mathrm{CI}_{95\%} = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1+\frac{z^2}{n}}. \tag{12}$$

5. Results

5.1. Model Comparison

Table 5 reports the HaluEval test results for eleven baselines and EC-XGB, averaged over three seeds where a seed applies. The F1 ranking is tight: the random forest is 0.0004 behind the conventional model, and EC-XGB is 0.0009 ahead. That lead is much smaller than the width of the conventional seed-42 bootstrap interval [0.9812, 0.9895], so the ordering is not resolved by the data. The benchmark is saturated, and the differences are small. The ranking columns tell the same story. EC-XGB is first among the feature-based models at AUROC 0.9984 and PR-AUC 0.9987, with the conventional model and random forest within 0.0006; RoBERTa is higher at 0.9990 and 0.9993. Calibration is separate: the conventional model has the lowest ECE, but its Brier score is higher than EC-XGB and RoBERTa, so the two summaries disagree. The weak controls are NLI-only and full-input TF-IDF at 0.6714 and 0.5999 F1; the answer-only TF-IDF model reaches 0.9224, and adding the question and context hurts. The reduced external ensemble reaches F1 0.9517 and AUROC 0.9850; the fixed-hp m3 model is higher by 0.0337 F1 [0.0262, 0.0419] and 0.0135 AUROC [0.0098, 0.0177], with ECE 0.0078 against 0.0141. The gap is consistent with the nonlinear learner and wider feature set, but the baseline is reduced and trained on fewer rows, so the comparison is descriptive.

The XGBoost margin over the random forest is 0.0004 F1, with McNemar $p = 0.502$; the difference is not significant. The margin over logistic regression is 0.0187 F1 ($p = 1.86 \times 10^{-10}$) and over the heuristic 0.0495 ($p = 2.28\times10^{-30}$). A source-only encoder wins in-domain (F1 0.9912 vs 0.9855) but is near chance on each RAGTruth task (AUROC 0.458 and 0.491 vs 0.582 for EC-XGB); pooling gives 0.337 because it mixes between-task score shifts with within-task ranking. An adapted encoder on the same multi-source rows reaches 0.694 on the seen tasks, above EC-XGB's 0.582, so the seen-task gain is not specific to the feature model. Neither model generalizes to unseen QA (0.490 vs 0.549), and FaithBench is worse for both (0.434 vs 0.594). Adaptation helps any model on seen tasks; the feature model keeps its relative ordering on unseen tasks.

Table 5
Test-set performance on HaluEval QA, mean over seeds 42/123/456 where a seed applies; deterministic rows are single fits (n = 3,000).

Model	P	R	F1	AUROC	PR-AUC	MCC	ECE	Brier
Heuristic (1−overlap)	0.9473	0.9233	0.9352	0.9132	0.8196	0.8723	0.3541	0.2672
Length-only (2 feats)	0.9454	0.9460	0.9457	0.9759	0.9722	0.8913	—	—
Logistic regression	0.9768	0.9553	0.9660	0.9932	0.9901	0.9329	0.0099	0.0253
Random forest	0.9897	0.9789	0.9842	0.9980	0.9983	0.9687	0.0143	0.0127
NLI-only	0.6479	0.6967	0.6714	0.7177	0.7531	0.3189	0.0678	0.2182
TF-IDF (answer-only)	0.9550	0.8920	0.9224	0.9696	0.9665	0.8519	0.0831	0.0662
TF-IDF (full input)	0.6397	0.5647	0.5999	0.6754	0.6785	0.2484	0.0679	0.2328
External ensemble[†]	0.9508	0.9527	0.9517	0.9850	0.9856	0.9033	0.0141	0.0391
XGBoost (conventional)	0.9932	0.9762	0.9846	0.9979	0.9983	0.9697	0.0045	0.0117
EC-XGB (m3)	0.9939	0.9773	0.9855	0.9984	0.9987	0.9715	0.0074	0.0108
RoBERTa-base	0.9955	0.9869	0.9912	0.9990	0.9993	0.9825	0.0070	0.0081
Majority (negative)	0.0000	0.0000	0.0000	—	—	0.0000	—	—

Note. ECE and Brier are computed on the raw score and measure the reliability of each risk score. The neural baseline is a fine-tuned RoBERTa-base with 256-token inputs. The majority baseline predicts the negative class for every row, so probability metrics are undefined. The EC-XGB row is the tuned m3 checkpoint; the m3 row in Table 6 is the fixed-hp refit. The conventional row is the B2 checkpoint; the m0 row in Table 6 uses the fixed configuration described there. The length-only row is a single deterministic logistic regression on log-scaled character and word counts, evaluated on the same test split at the 0.5 threshold, and its ECE and Brier were not computed. [†]The external ensemble is a reduced reimplementation of the lightweight feature ensemble of Sundaragiri et al. [10]: five features (ROUGE-L, SBERT cosine, two NLI probabilities, answer length) and a logistic regression, fit on the 14,000 HaluEval training rows, with the threshold fixed at 0.5; a validation-selected threshold of 0.55 gives F1 0.9514, so the choice does not change the comparison. It is not the original authors' model or reported number.

Figure 4 shows the curves behind Table 5. In HaluEval the feature-based models overlap near the top-left corner because the benchmark is saturated; in RAGTruth EC-XGB is above the conventional model on summarization and data-to-text, and both are close to the diagonal on QA. The curves use seed 42; the table values are three-seed means.

5.2. EC-XGB and Ablation

Table 6 reports the component ablation under one fixed configuration (max depth 5, learning rate 0.05, 300 trees, subsample 0.7, colsample 0.7, selected once on the m3 seed-42 run), so the columns measure feature effects rather than tuning effects. The in-domain F1 span is only 0.0006 across m0–m3. The claim aggregates change F1 by −0.0004 at m2 relative to m1, and the full m3 stack changes it by +0.0001; neither is an aggregate gain. On the pooled RAGTruth split, AUROC rises from 0.494 to 0.590 and raw ECE falls from 0.635 to 0.275; on held-out QA the comparison is 0.556 vs 0.558, within seed noise. McNemar gives $p = 0.789$ for m0 vs m3, so the ablation does not resolve a difference. The gain is not from the claim features; it appears when the multi-source rows and source indicator are added. The MedHallu source-zero check suggests the indicator is not the main driver, but a seen-task source ablation would settle it. The deployed checkpoint reaches 0.9855 under its own tuned configuration, 0.0001 above the fixed ablation.

Table 7 repeats the ablation on the fixed-hp m3 model. Lexical overlap is the largest contributor (−0.0171 F1, −0.0040 AUROC), length second (−0.0066, −0.0012), and NLI third (−0.0040). The claim block costs 0.0010 F1, about the seed standard deviation, so an aggregate claim gain is not resolved; entity, numeric, hedging, semantic, and source each change F1 by less than 0.0011. Lexical overlap and length therefore carry the in-domain score, and the remaining groups add ranking information without moving F1 far.

Figure 5 shows the seed-level spread behind Table 6. Each point is one seed, and the black bar is the mean. The F1 panel spans 0.9845 to 0.9856 across the four variants, and the three seeds within a variant differ by at most 0.0007. The AUROC panel is tighter, with a within-variant spread below 0.0005. The ECE panel is the noisiest. The m0 variant sits at 0.0047 to 0.0054 with a standard deviation of 0.0004, while m2 and m3 range from 0.0032 to 0.0096 with standard deviations above 0.0016. The mean ordering is m3 > m1 > m2 > m0, but the per-seed ranks flip and the differences are below seed noise. The ECE sign is stable: m0 is lower than m3 on every seed, by 0.0014 to 0.0048, so the ablation does not establish a calibration advantage for m3.

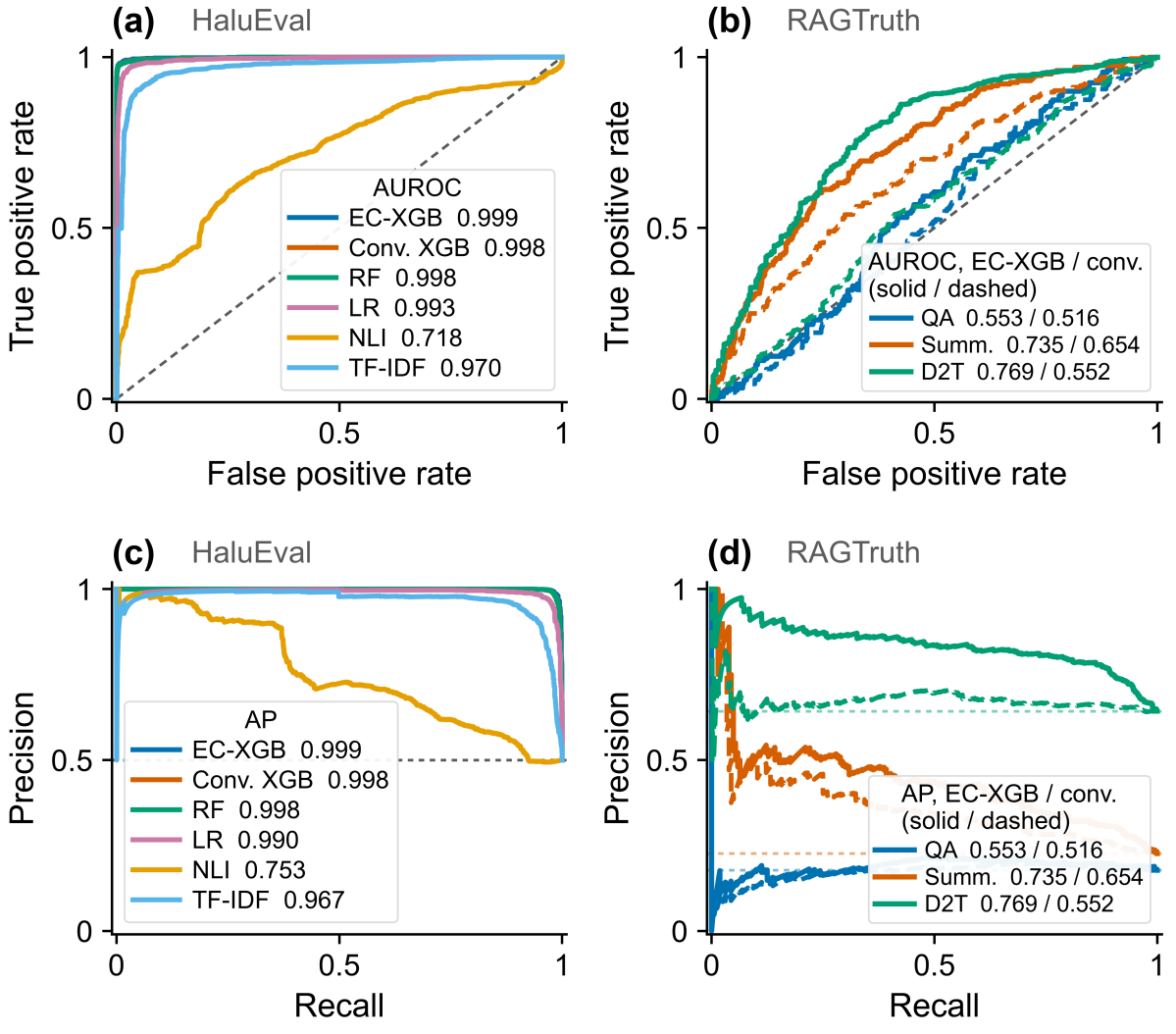


Figure 4: ROC and precision-recall curves on HaluEval and RAGTruth (seed 42).

Table 6
EC-XGB component ablation on the HaluEval test set with one fixed hyperparameter configuration for every variant, mean over three seeds (n = 3,000 for HaluEval columns; n = 2,700 for RAGTruth columns).

Variant	Features	F1	AUROC	MCC	ECE	RAGTruth AUROC	RAGTruth ECE
m0 conventional	26	0.9849	0.9979	0.9701	0.0050	0.4937	0.6352
m1 debias	26	0.9853	0.9976	0.9710	0.0042	0.5567	0.6260
m2 + claims	34	0.9849	0.9985	0.9701	0.0051	0.5461	0.6359
EC-XGB (m3)	35	0.9854	0.9986	0.9713	0.0078	0.5898	0.2746

Note. ECE and Brier are computed on the raw score. All variants use max depth 5, learning rate 0.05, 300 trees, subsample 0.7 and colsample 0.7, selected once on the m3 seed-42 run; no variant is tuned separately. m0 is the conventional reference. m1 replaces raw length counts with log-scaled values and adds monotone constraints (26 features). m2 adds the eight claim aggregates (34 features). m3 adds the RAGTruth non-QA training rows and a source indicator (35 features). The last two columns use the pooled RAGTruth test split (2,700 rows).

Table 7
Feature-group ablation of the fixed-hyperparameter EC-XGB m3 model on the HaluEval test set, mean over seeds 42/123/456 (reference F1 0.9861, AUROC 0.9985, MCC 0.9726; n = 3,000).

Removed group	Features	ΔF1	ΔAUROC	ΔMCC
length	4	−0.0066	−0.0012	−0.0127
lexical	4	−0.0171	−0.0040	−0.0343
entity	4	−0.0010	+0.0000	−0.0020
nli	6	−0.0040	−0.0004	−0.0080
numeric	4	−0.0003	+0.0001	−0.0007
hedging	2	+0.0000	+0.0001	0.0000
semantic	2	0.0000	+0.0002	+0.0000
claims	8	−0.0010	−0.0003	−0.0020
source	1	−0.0001	+0.0001	−0.0002

Note. Deltas are relative to the full 35-feature m3 model refit under the same fixed configuration (max depth 5, learning rate 0.05, 300 trees, subsample 0.7, colsample 0.7). Every variant is trained on the same 24,056 rows and evaluated once on the 3,000 test rows. No variant is tuned or early-stopped. The reference differs from Table 6 by 0.0007 F1 because the feature matrix is rebuilt for this run; the difference is within the seed standard deviation of the refit (0.0010).

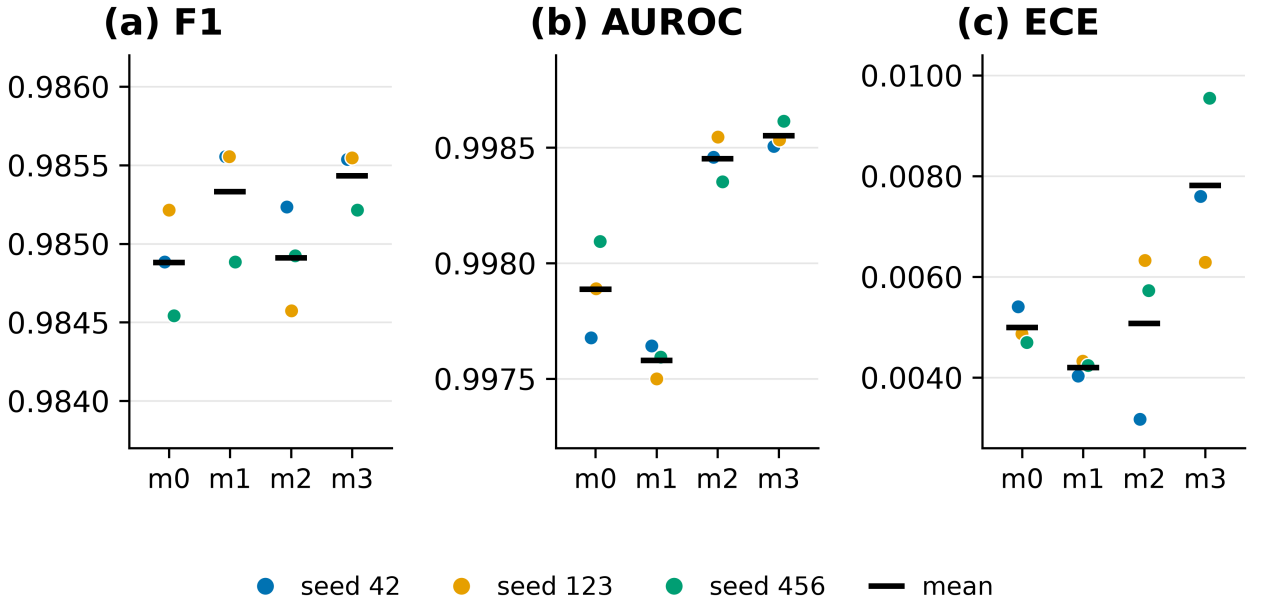


Figure 5: Per-seed F1, AUROC and ECE for the fixed-hyperparameter B6 variants.

5.3. Calibration, Shift, and External Validity

The raw, Platt-scaled, and isotonic versions of the conventional model give ECE 0.0045, 0.0091, and 0.0070 on HaluEval, so the calibrated ECEs are not lower in this sample. Figure 6 shows why. On HaluEval, the points at the two ends sit on the diagonal; the middle of the score range holds few distinct values, so the curve has only a few stable bins. The Platt and isotonic curves overlap the raw curve at both ends. The m3 raw score is also well calibrated in-domain, with ECE 0.0078, so there is little room for calibration to improve on HaluEval.

Platt scaling fits a two-parameter logistic map from the raw score s to a probability,

$$\hat{p} = \sigma(as + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{13}$$

where a is the calibration slope and b is the intercept. A slope below one means the raw score is overconfident, and a slope near one means the scale is already usable. The Brier skill score compares the calibrated score with a constant

predictor at the evaluation prevalence,

$$\mathrm{BSS} = 1 - \frac{\mathrm{BS}}{\mathrm{BS}_{\mathrm{prior}}}. \tag{14}$$

A positive BSS means the score beats the prior, and a non-positive BSS means it does not.

For the conventional model, the RAGTruth QA histogram in Figure 6 shows almost every row above 0.99, at a mean confidence of 0.998 against an accuracy of 0.178. Source Platt leaves the confidence near 0.99. The m3 score behaves the same way. The fixed-hp m3 raw score is severely overconfident: ECE 0.731 and Brier 0.689, with a Cox slope of 0.18 on the 5,034 calibration rows. Source Platt cannot repair a base-rate shift. Target Platt lowers ECE to 0.125 and Brier to 0.161, but the Brier skill score against the test prior is −0.10. A constant predictor at the test prevalence has Brier 0.146, lower than every calibrated row, and flags no rows at 0.5. Calibration fixes the scale, not the ranking; AUROC stays at 0.558. The conventional target-Platt row behaves the same way at ECE 0.134 and Brier 0.164. The label budget is the practical constraint, and it is a distribution rather than a point.

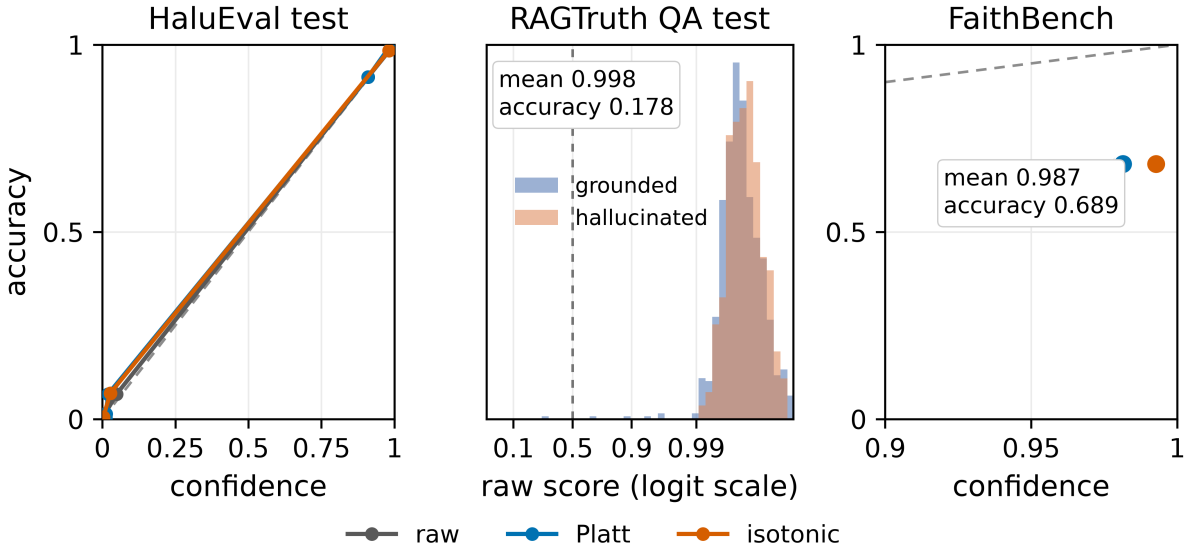


Figure 6: Calibration of the B2 conventional model. Left and right: quantile-bin reliability diagrams for HaluEval and FaithBench (FaithBench is aggregated because its scores concentrate near 1.0). Middle: RAGTruth QA raw-score distribution on a logit scale; both classes pile near 1.0.

The source-only operating point (P0) selects the threshold on the HaluEval validation split at a 5% false-positive budget. On the HaluEval test split, recall there is 99.0–99.5% and the false-positive rate stays near 5%. On RAGTruth QA the same source threshold flags 97.4% of the tuned checkpoint, so P0 is not a usable target operating point. Table 8 reports both models on the same 900-row RAGTruth QA test set, whose prevalence is 17.8%. The calibration split has 31.1% positives. A prevalence-only predictor set to the test prevalence, 0.1778, shows what a constant score can achieve. On the test rows it reaches ECE 0.000 and Brier 0.146, and it predicts no positives at the 0.5 threshold. At the calibration prevalence of 0.3107 the same predictor gives ECE 0.133 and Brier 0.164, which matches the conventional target-Platt row. The calibrator therefore sets the level to the calibration prevalence, and the residual ECE is dominated by the prevalence gap. At the test prevalence of 0.1778 a constant predictor has Brier 0.146, lower than the target-calibrated rows (0.161–0.164). Figure 7 shows the raw scores and the target-recalibrated scores for the conventional model. Calibration fixes the scale, not the ranking.

The conventional target-thresholded mode (P2) selects a threshold at a 5% false-positive budget (0.313) on the calibration split and reaches a test recall of 5.0% at a false-positive rate of 5.8% (precision 15.7%, F1 0.076, Wilson 95% intervals: recall [2.6, 9.6], FPR [4.3, 7.7]). The test rate exceeds the 5% target because the threshold is fixed on calibration. At 17.8% prevalence, random flagging at a 5.8% false-positive rate yields 5.8% recall, so the P2 result is chance-level.

Table 8
Calibration of the fixed-hp m3 model and the conventional target-Platt reference on the RAGTruth QA test set (n = 900, 17.8% positive).

Predictor	Fit	ECE	Brier	BSS	Slope	Flagged
Prior (test prevalence)	constant 0.1778	0.000	0.146	0.000	—	0.0%
Prior (cal prevalence)	constant 0.3107	0.133	0.164	−0.12	—	0.0%
m3 raw	none	0.731	0.689	−3.71	0.18 [0.12, 0.27]	96.8%
m3 source Platt	HaluEval val	0.765	0.742	−4.07	—	96.8%
m3 target Platt	RAGTruth cal	0.125	0.161	−0.10	—	0.0%
Conventional target Platt	RAGTruth cal	0.134	0.164	−0.12	—	0.0%

Note. ECE and Brier are on the 900 test rows. BSS uses the test-prevalence prior as the reference. The slope is the Cox slope of the raw score on the 5,034 calibration rows; a value of one is calibrated, and the brackets are the envelope of the three per-seed 95% bootstrap CIs. Row-bootstrap 95% CIs for the headline rows are raw ECE [0.706, 0.757], raw Brier [0.664, 0.713], target ECE [0.099, 0.151], and target Brier [0.151, 0.171]. Platt fits use validation or calibration rows only. The test prior row is zero by construction; the calibration-prior row is evaluated on the test rows. The conventional row uses the B2 target-Platt checkpoint and is shown as the reference behind the P2 analysis.

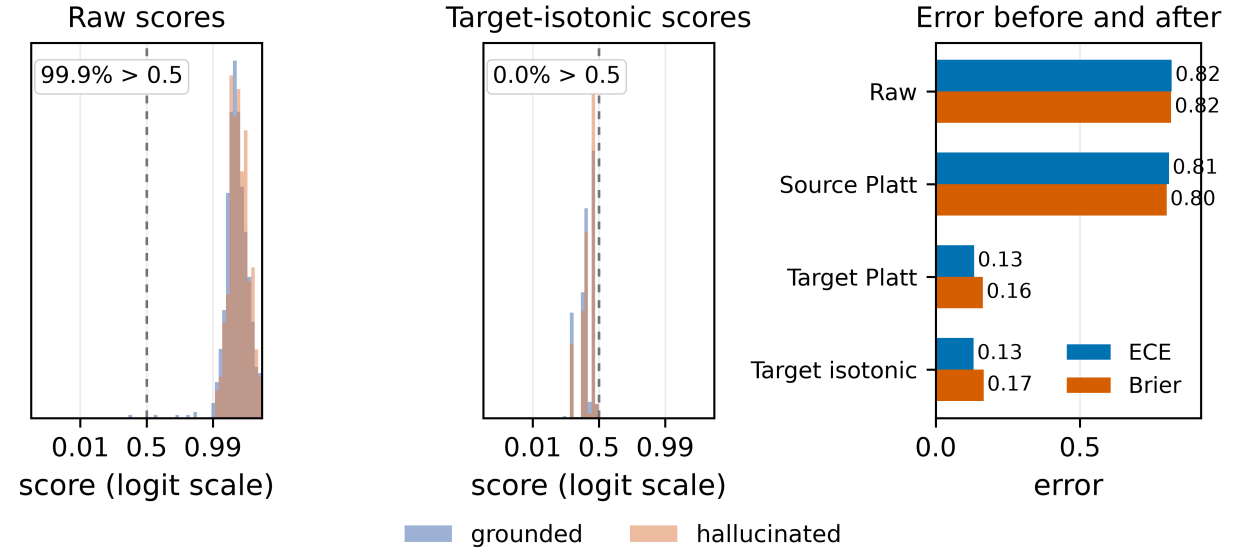


Figure 7: Target-domain recalibration of the conventional model on RAGTruth QA. Left and middle: raw and target-isotonic score distributions on a logit scale. Right: ECE and Brier for raw, source Platt, target Platt, and target isotonic.

At the 0.5 threshold, P1 flags no RAGTruth QA rows, and the maximum target-Platt score (0.313) is below the API's lowest risk band (0.35); the deployed card therefore relies on the per-claim verdict adjustment, not on the pre-verdict score alone. All calibration metrics in this section refer to the pre-verdict display score. The card value is adjusted by claim verdicts and is not covered by these ECEs; the displayed percentage is a design aid, not a validated probability.

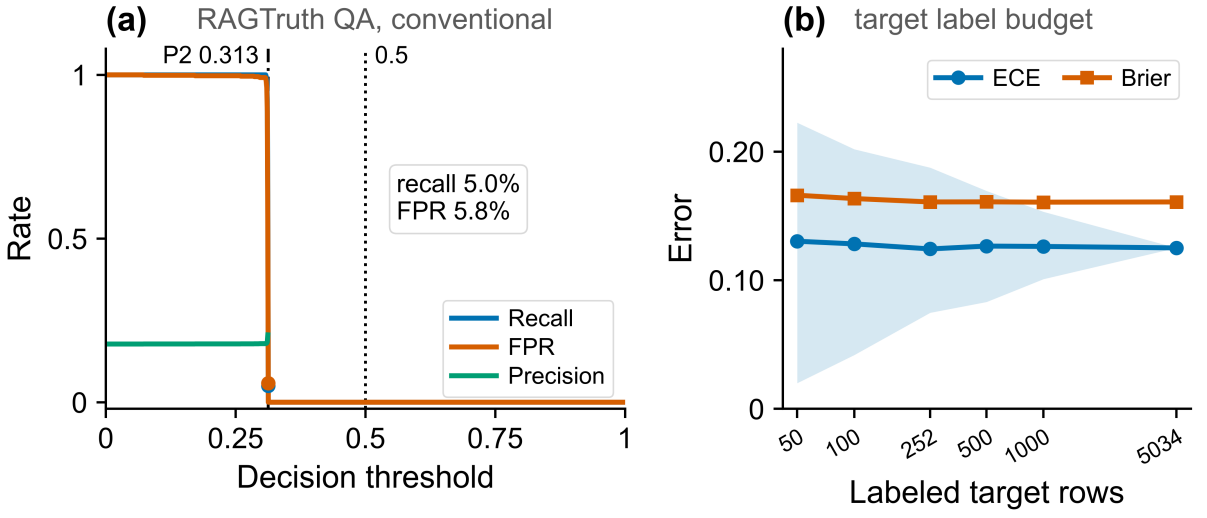


Figure 8: Operating point for the conventional target-Platt model and 100-draw label budget for the fixed-hp m3 calibrator on RAGTruth QA.

Figure 8 shows the operating point and the label budget. Panel (a) sweeps the threshold for the conventional target-Platt model on RAGTruth QA, and the vertical line marks the selected threshold at 0.313, which is the P2 row in Table 8. A lower threshold raises recall and raises the false-positive rate. Panel (b) plots 100 draws at each budget. The mean ECE is 0.130 at 50 rows, 0.124 at 252 rows, and 0.125 at 5,034 rows, but the 95% band narrows from [0.020, 0.223] at 50 rows to [0.075, 0.188] at 252 rows and collapses at the full split. A single 252-row draw can therefore miss the full-split ECE by about 0.06, so the budget saturates in the mean but not in a single draw. A small labeled target sample sets the scale only when its draw is checked against a held-out split.

Table 9 reports the coverage. Split conformal prediction adds a coverage guarantee to the score. The nonconformity score for label y is $s(x, y) = 1 - \hat{p}_y(x)$, where $\hat{p}_1 = \hat{p}$ and $\hat{p}_0 = 1 - \hat{p}$. The calibration quantile and the prediction set are

$$\hat{q} = \text{Quantile}_{1-\alpha}\left(\{s(x_i, y_i)\}_{i=1}^{n}\right), \qquad C(x) = \{y : s(x, y) \le \hat{q}\}. \tag{15}$$

A class-conditional version replaces the single quantile with one quantile per label,

$$\hat{q}_y = \text{Quantile}_{1-\alpha}\left(\{s(x_i, y_i) : y_i = y\}\right), \qquad C(x) = \{y : s(x, y) \le \hat{q}_y\}. \tag{16}$$

The main run fits the Platt map and the quantile on the same 5,034 rows, so its calibration coverage is a construction check. A disjoint sensitivity fits Platt on 2,514 rows and the quantile on 2,520 rows. On that split, the 900 QA test rows reach 0.944 coverage at 90%, so the marginal guarantee holds. It hides the class split: positive coverage is 0.685, while negative coverage is 1.000. A class-conditional quantile raises positive coverage to 0.887, still slightly below nominal. FaithBench coverage falls to 0.320 and positive coverage to 0.003, because a Platt map fitted on QA sends almost every FaithBench score to the negative side. Conformal prediction therefore certifies the marginal rate only when the calibrator and the quantile are fitted on disjoint rows, and it does not transfer across the task shift. The per-class rate is the one a reviewer needs when positives are 17.8% of the target.

Multicalibration by answer length and claim count shows where the average ECE hides error. On the 900 QA test rows, the target-Platt score has overall ECE 0.125. The worst length bin is 60 words or fewer (n=154), with ECE 0.275; the worst claim-count bin is a single claim (n=36), with ECE 0.346. The Brier score does not follow the ECE

Table 9
Split and class-conditional conformal coverage on the 900 RAGTruth QA test rows and 750 FaithBench rows, mean over seeds 42/123/456. The Platt map and the conformal quantile are fitted on disjoint halves of the 5,034 QA calibration rows.

Dataset	Method	α	Coverage	Pos.	Neg.	Set size
QA test	split	0.10	0.944	0.685	1.000	1.59
QA test	split	0.05	0.972	0.842	1.000	1.75
QA test	class-cond.	0.10	0.886	0.887	0.886	1.69
QA test	class-cond.	0.05	0.936	0.925	0.938	1.80
FaithBench	split	0.10	0.320	0.003	1.000	1.00
FaithBench	split	0.05	0.323	0.006	1.000	1.01

Note. Coverage is the share of rows whose true label is in the prediction set. Pos. and Neg. are the same rate computed within each class. Set size is the mean number of labels in the set, with a maximum of two. The class-conditional rows use a separate quantile per label on QA. FaithBench is a zero-shot summarization corpus and is never used for fitting. The in-sample run, which fits Platt and the quantile on the same rows, gives 0.940 and 0.967 coverage on the QA test rows and 0.900 and 0.950 on the calibration rows; the latter is a construction check and is not reported as the primary result.

ranking: it is lower in the short and one-claim bins, so the two summaries disagree about where the calibration error is concentrated. The average is therefore not a safe summary for short answers or one-claim answers, which are the cases where a reviewer most needs a reliable number. FaithBench is not improved by the QA calibrator: its target-Platt ECE is 0.629 against 0.453 for the raw score, so the calibrator is domain-specific.

The conventional reference m0 is trained only on HaluEval, whereas EC-XGB also sees RAGTruth summarization and data-to-text rows, so those two pairs measure adaptation, not general shift (Table 10). On the pooled RAGTruth test split, the flag rate falls from 99.9% to 61.3% and AUROC rises from 0.475 to 0.582. The gain is task-specific: summarization AUROC rises from 0.634 to 0.736 and data-to-text from 0.567 to 0.771. Held-out QA remains near chance (AUROC 0.547 to 0.549). FaithBench AUROC rises from 0.509 to 0.594, but F1 falls from 0.813 to 0.120 and raw ECE rises from 0.292 to 0.450, so the raw scores transfer poorly. The conventional F1 matches the all-positive baseline because it flags almost every row.

Table 11 reports the three external corpora and the task-shift pair. MedHallu is the strongest evidence that the ranking survives a domain change and the clearest evidence that the probability does not. The fixed-hp m3 model ranks the paired hallucinated answer above the grounded answer in 86.7% of 1,000 medical pairs [84.5, 88.7], with AUROC 0.821 [0.806, 0.837] on the 2,000 rows. The pair accuracy over M pairs is

$$\mathrm{PA} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\left[\hat{p}(a_m^+) > \hat{p}(a_m^-)\right], \tag{17}$$

where a_m^+ is the hallucinated answer and a_m^- the grounded answer. The conventional m0 model reaches 82.2% and 0.781. The length-only baseline is inverted, at 20.0% and 0.260, because grounded MedHallu answers are longer than the paired hallucinations (39.7 vs 27.8 words), the opposite of HaluEval (2.2 vs 11.0). The ranking therefore transfers, but the scale does not: m3 flags every row, so its F1 is the all-positive 0.667 and its ECE is 0.493. The source indicator is one for MedHallu and zero for HaluBench; a source-zero sensitivity on MedHallu changes pair accuracy by at most 0.003 and AUROC by at most 0.002, so the indicator does not drive the result. HaluBench is a 30% stratified sample of the 3,000 non-overlapping rows. Its overall AUROC is 0.600, with covidQA 0.833, DROP 0.513, and FinanceBench 0.527; the model transfers to the biomedical subset but is at chance on the other two. The length-only baseline is at 0.514 overall, so 0.600 is not a length artifact.

The task-shift test is the weakest. On dialogue, m3 flags 99.3% of rows and its F1 equals the all-positive baseline, with AUROC 0.763; the length-only baseline reaches 0.724. On summarization, m3 reaches AUROC 0.717 and flags 37.5%, while the length-only baseline reaches 0.788 and flags every row. The conventional m0 model flags 99.2% of dialogue rows and 97.7% of summarization rows, with AUROC 0.741 and 0.605. On dialogue, both models collapse to the all-positive baseline in F1, though their AUROCs stay above chance; on summarization, both rank below the length-only model. This is a task-shift test, not a second independent QA corpus.

An index stress test asks whether the weak held-out QA result is a retrieval problem. The hybrid BM25 and FAISS index covers the 989 RAGTruth QA source documents. On a 300-row, 50-group sample, recall@1 is 0.94, recall@5

Table 10
External and adaptation performance of B6 m0 and EC-XGB m3 by dataset and task on the held-out RAGTruth test rows and FaithBench, mean over three seeds.

Dataset / task	Model	F1	AUROC	PR-AUC	Flagged	ECE
RAGTruth pooled	Conventional (m0)	0.5181	0.4753	0.3139	99.9%	0.6346
RAGTruth pooled	EC-XGB (m3)	0.5608	0.5815	0.3588	61.3%	0.2782
RAGTruth QA	Conventional (m0)	0.3022	0.5465	0.1911	99.9%	0.8149
RAGTruth QA	EC-XGB (m3)	0.3081	0.5491	0.1884	97.4%	0.7433
RAGTruth summarization	Conventional (m0)	0.3702	0.6343	0.3536	99.8%	0.7499
RAGTruth summarization	EC-XGB (m3)	0.3434	0.7358	0.4449	11.6%	0.0519
RAGTruth data-to-text	Conventional (m0)	0.7830	0.5665	0.6831	100.0%	0.3390
RAGTruth data-to-text	EC-XGB (m3)	0.8234	0.7705	0.8374	74.9%	0.0527
FaithBench	Conventional (m0)	0.8130	0.5089	0.6735	99.5%	0.2917
FaithBench	EC-XGB (m3)	0.1201	0.5941	0.7109	7.3%	0.4500

Note. ECE is computed on raw scores, not on the target-calibrated display score. The conventional rows are B6 m0, a separate 26-feature tuning run under the EC-XGB protocol, and the EC-XGB rows are m3. Every row is a three-seed mean. m0 is trained on HaluEval only, so its external rows are zero-shot. EC-XGB adds the RAGTruth summarization and data-to-text training rows, so those two pairs measure adaptation, and RAGTruth QA is held out. F1 uses the fixed 0.5 threshold, and Flagged is the share of rows above it. RAGTruth pooled uses the 2,700 held-out test rows, each RAGTruth task uses 900, and FaithBench uses 750. The m0 F1 matches the all-positive baseline to within 0.003 on QA (0.302), pooled (0.518), and FaithBench (0.810) because it flags almost every row. These rows use the frozen tuned checkpoint; the fixed-hp judge rows give 96.8% and AUROC 0.558 on QA, within 0.01 of the tuned values.

Table 11
External validity on corpora that are never used for training, calibration, threshold selection, or model choice (scores averaged over seeds 42/123/456). Pair accuracy is the share of the 1,000 MedHallu pairs in which the hallucinated answer scores above the grounded one.

Dataset (n)	Model	F1	AUROC	ECE	Flagged	Pair acc.
MedHallu (2,000)	EC-XGB (m3)	0.667	0.821	0.493	100.0%	0.867
	Conventional (m0)	0.667	0.781	0.499	100.0%	0.822
	Length-only	0.667	0.260	0.500	100.0%	0.200
HaluBench (900)	EC-XGB (m3)	0.588	0.600	0.334	50.3%	—
	Conventional (m0)	0.585	0.609	0.346	51.6%	—
	Length-only	0.441	0.514	0.420	33.2%	—
HaluEval dialogue (2,000)	EC-XGB (m3)	0.670	0.763	0.488	99.3%	—
	Length-only	0.680	0.724	0.438	95.9%	—
HaluEval summarization (2,000)	EC-XGB (m3)	0.583	0.717	0.113	37.5%	—
	Length-only	0.667	0.788	0.500	100.0%	—

Note. MedHallu is medical QA; HaluBench is a non-overlapping subset from DROP, FinanceBench, and covidQA (300 rows each); HaluEval dialogue and summarization are task-shift tests within the HaluEval family, not a second independent QA corpus. The length-only baseline is a logistic regression on log answer length, fit once on HaluEval train only. HaluBench rows are not paired, so pair accuracy is undefined. All evaluation sets are balanced 50/50 by construction, so F1 and flag rate are prevalence-dependent and do not describe natural deployment. Flagged is the share of rows above 0.5. The feature models use the fixed-hp configuration; no external row is used for tuning.

and recall@10 are 1.00, and MRR is 0.959. Replacing the gold context with the top-5 retrieved passages changes AUROC from 0.545 to 0.521, F1 from 0.345 to 0.349, and ECE from 0.706 to 0.704. The AUROC delta is −0.023 [−0.061, 0.011], which includes zero. Retrieval is therefore not the bottleneck: the model is near chance on RAGTruth QA even with the gold source passage. The qrels are the known source passage, not human web-search judgments, so this measures index recall, not end-to-end search quality. The sample has 50 source groups, so the interval is wide.

Figure 9 shows the raw score distributions of the conventional model on HaluEval and on the external test rows. The first panel is the HaluEval test set, where the two classes separate into two clusters at the ends of the range. The remaining panels are the held-out RAGTruth test rows (900 per task, 2,700 pooled) and FaithBench. In each external panel, both classes pile against 1.0, so the raw score no longer separates hallucinated from correct answers. The 0.5 threshold therefore flags almost every row. This is the visual form of the 99.9% flag rate in Table 10.

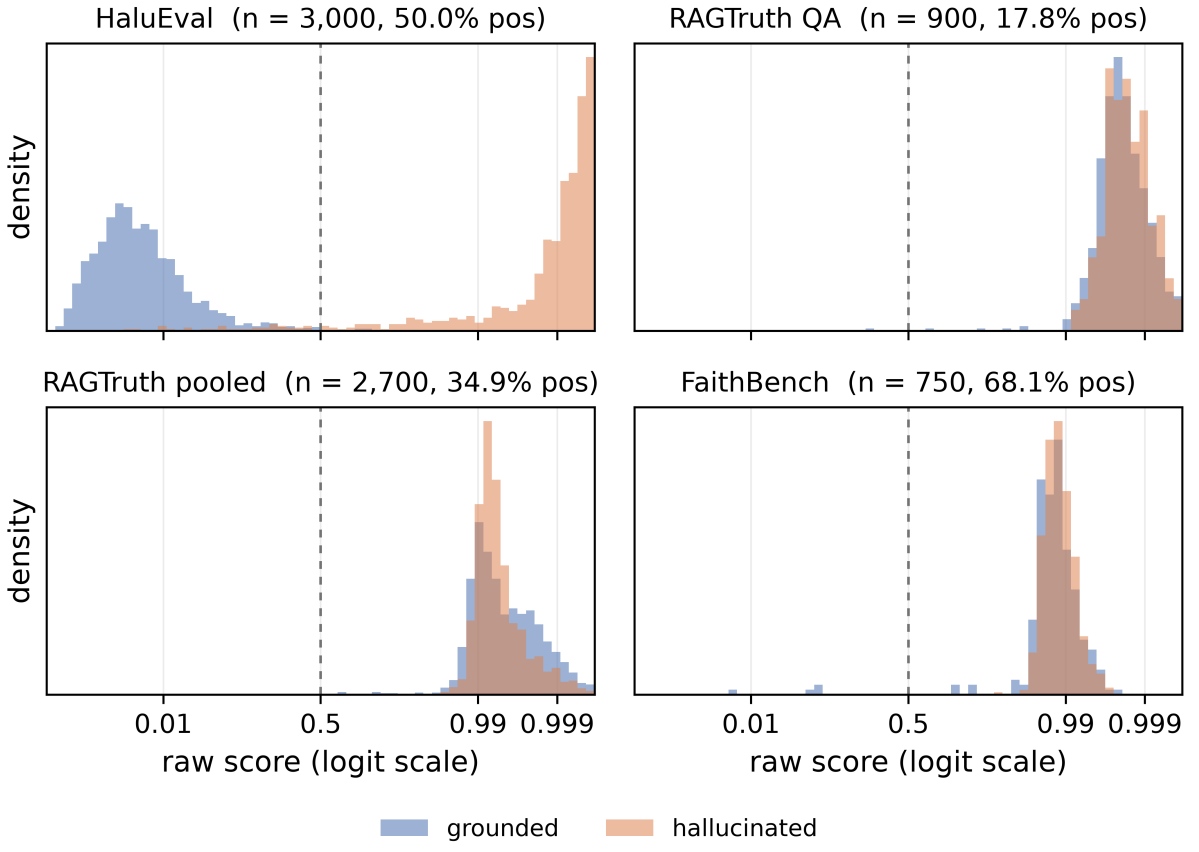


Figure 9: Raw score distributions of the conventional B2 model on HaluEval and the held-out external test rows (RAGTruth QA, pooled RAGTruth, FaithBench), on a logit scale. Both classes pile near 1.0 outside HaluEval.

5.4. Statistical Analysis

Statistical tests use the paired fixed-hyperparameter predictions and 2,000 two-level bootstrap draws, where groups are resampled within each seed and then seeds are resampled. McNemar's test compares the discordant counts b and c on the paired seed-42 rows, where b is the number of rows where the first model is right and the second is wrong, and c is the reverse. The test statistic is

$$\chi^2 = \frac{(|b-c|-1)^2}{b+c}, \tag{18}$$

with a continuity correction. Holm's method orders the m p-values and adjusts the i-th value as

$$p_{(i)}^{\text{Holm}} = \min_{j \geq i} \left\{ \min\left(1, (m-j+1)p_{(j)}\right) \right\}, \tag{19}$$

which controls the family-wise error rate across the four baseline comparisons. The paired bootstrap delta is

$$\Delta_b = \theta_{\mathrm{EC}}(\mathcal{D}_b) - \theta_{\mathrm{conv}}(\mathcal{D}_b), \tag{20}$$

where $\mathcal{D}_b$ is a resample of groups and θ is the metric. The 95% interval is the 2.5th and 97.5th percentiles of the Δ_b,

$$\mathrm{CI}_{95\%} = \left[\Delta_{(\alpha/2)}, \Delta_{(1-\alpha/2)}\right]. \tag{21}$$

The two-level version resamples groups within each seed and then resamples seeds,

$$\Delta^* = \frac{1}{S}\sum_{s=1}^{S}\left[\theta^*_{\mathrm{EC}}(\mathcal{D}^*_s) - \theta^*_{\mathrm{conv}}(\mathcal{D}^*_s)\right], \tag{22}$$

where $\mathcal{D}^*_s$ is a group resample for seed s and $S = 3$. The Wilcoxon signed-rank statistic is

$$W = \sum_{s=1}^{S}\mathrm{rank}(|d_s|)\mathbf{1}[d_s > 0], \tag{23}$$

where d_s is the difference between the two models on seed s and $S = 3$.

Table 12 gives the McNemar counts. EC-XGB and the conventional model are not significantly different under the deployed checkpoints ($p = 0.45$), and the fixed-hyperparameter ablation gives $p = 0.789$ on the same test rows. That is the primary comparison, and it is not part of the Holm family, which covers the four baseline comparisons against the conventional model. The random forest matches the conventional model ($p = 0.502$). Logistic regression and the heuristic are significantly worse ($p = 1.86 \times 10^{-10}$ and $p = 2.28 \times 10^{-30}$), and both survive Holm correction. The Wilcoxon test across three seeds cannot reach significance (minimum $p = 0.25$). Table 13 reports the two-level bootstrap. On HaluEval, the m3−m0 F1 delta is +0.0006 [−0.0010, 0.0021], which includes zero, and the AUROC delta is +0.0007 [0.0002, 0.0012], which excludes zero by a small margin. The m2−m1 F1 delta is −0.0004 [−0.0020, 0.0012], and the m3−m1 delta is +0.0001 [−0.0015, 0.0018], so the claim features do not add an aggregate gain. Their AUROC deltas are +0.0009 and +0.0010, both with intervals above zero, so the added features reorder a few rows without changing F1.

On held-out RAGTruth QA the m3−m0 F1 delta is +0.0076 [0.0045, 0.0109], but the AUROC delta is +0.0017 [−0.0270, 0.0291], which includes zero. The F1 gain comes from flagging almost every row, not from better ranking. On the pooled seen tasks the F1 delta is +0.0451 [0.0348, 0.0554] and the AUROC delta is +0.0962 [0.0644, 0.1295], which is the adaptation gain. The paired fixed-minus-tuned contrasts all include zero, so the rebuild does not change these conclusions.

Table 12
McNemar tests on the paired seed-42 HaluEval predictions.

Comparison	b	c	p	Holm p
EC-XGB vs conventional	10	6	0.453	—
EC-XGB vs conventional (fixed hp)	6	8	0.789	—
Conventional vs random forest	12	8	0.502	0.502
Conventional vs logistic regression	69	11	1.86×10^{-10}	3.7×10^{-10}
Conventional vs heuristic	158	9	2.28×10^{-30}	6.8×10^{-30}
Conventional vs NLI-only	1006	26	5.6×10^{-204}	2.2×10^{-203}

Note. b is the number of rows where the first model is right and the second is wrong. c is the reverse. The p values use the chi-square form with continuity correction. Holm values cover the four baseline comparisons against the conventional model. The fixed-hp row compares m0 and m3 under the single configuration of Table 6.

5.5. Explanation, Efficiency, and Judges

The explanation results use the fixed-hyperparameter m3 seed-42 checkpoint. Figure 10 ranks the top-15 features by mean absolute SHAP over 1,000 stratified HaluEval test rows. Context overlap dominates, followed by word count,

Table 13
Two-level bootstrap deltas for the fixed-hyperparameter ablation (2,000 draws; groups within seed, then seeds). Intervals are 95%.

Dataset	Comparison	ΔF1 [95% CI]	ΔAUROC [95% CI]
HaluEval	m3−m0	+0.0006 [−0.0010, 0.0021]	+0.0007 [0.0002, 0.0012]
HaluEval	m2−m1	−0.0004 [−0.0020, 0.0012]	+0.0009 [0.0004, 0.0014]
HaluEval	m3−m1	+0.0001 [−0.0015, 0.0018]	+0.0010 [0.0004, 0.0016]
RAGTruth QA	m3−m0	+0.0076 [0.0045, 0.0109]	+0.0017 [−0.0270, 0.0291]
RAGTruth pooled	m3−m0	+0.0451 [0.0348, 0.0554]	+0.0962 [0.0644, 0.1295]

Note. HaluEval groups are item pairs (1,500 groups); RAGTruth QA has 150 source groups and the pooled split has 450. The QA AUROC interval includes zero, so the held-out transfer result is not resolved. The pooled row measures adaptation to task types seen in training, not unseen-task transfer.

question-answer cosine, character count, and question-answer Jaccard; the claim-contradiction mean enters at rank eight. The top ten features carry 77.9% of the total attribution mass, and exact TreeSHAP reproduces the model margin to 9.2×10^{-6}. The m3 ranking agrees with the frozen B2 ranking (Spearman $\rho = 0.878$, Kendall $\tau = 0.730$), so the claim features refine the explanation rather than replace the original drivers. Attribution stability across seeds is high: against seed 42, Spearman is 0.972 for seed 123 and 0.950 for seed 456, top-10 overlap is nine of ten, and the top-5 set is identical. The explanation is stable to the seed, although the perturbation pilot below shows that it is less stable to content edits. The attributions are on the raw margin, not the display score. The audit cases are error-enriched, and their roles change under m3; they are discussed below.

Table 14 shows the m3 pilot. Content edits move the score (entity replacement, support removal, number replacement, date replacement), while structural edits barely move it (irrelevant insert 0.003, clause shuffle 0.0001). The top-1 feature flips in 54.5% of entity perturbations and in none of the other operators, and Spearman correlation stays between 0.73 and 0.999. The entity flip rate is higher than in the B2 run, so the m3 explanation is less stable when the answer entity changes. The pilot has between two and twenty changed cases per operator, so these values are descriptive, not effect estimates.

Table 14
Perturbation stability of the m3 raw score and SHAP ranking on HaluEval (seed 42, 20-case pilot).

Operator	n	Δscore	Large rate	Top-1 flip	Spearman
Entity replacement	11	0.339	0.364	0.545	0.848
Support removal	10	0.464	0.500	0.000	0.808
Number replacement	3	0.332	0.333	0.000	0.731
Date replacement	2	0.482	0.500	0.000	0.786
Irrelevant insert	20	0.003	0.000	0.000	0.922
Clause shuffle	2	0.000	0.000	0.000	0.999

Note. n counts perturbations that changed the feature vector. The pilot covers 20 cases and 120 attempts, of which 48 changed. Per-operator n is small, so the deltas are descriptive. Δscore is the mean absolute change in the m3 raw score.

A structured AI-assisted audit assessed 26 cases that were selected under B2 and re-scored under m3. Under m3 the same cases redistribute to 7 false positives, 12 false negatives, 3 borderline cases, 3 other cases, and 1 true positive. Each explanation was checked against the answer-context evidence, not the dataset label. Thirteen of the 26 explanations were judged implausible, three unsure, and ten plausible. The explanations are diagnostic aids, not faithful attributions. False positives inherit the length confound, and false negatives often reuse entities from the context. Two cases (2251 and 2436) appear to be label noise. Because the sample is error-enriched, the percentages describe the audit, not the test set. A three-author blinded review with AI pre-annotation covered this sheet and a 60-item overlap. The authors reached Fleiss $\kappa = 0.87$ for taxonomy, 0.94 for explanation, and 0.76 for claims, and agreed with the original pre-labels on 94%, 98%, and 88% of the 250 rows, so the labels are author-verified.

Table 15 gives the component latency. NLI and NER dominate the base median; XGBoost and SHAP are negligible. Adding the eight claim features raises the m3 path to 127.5 ms at the median, with a p95 of 234.6 ms; claim NLI alone

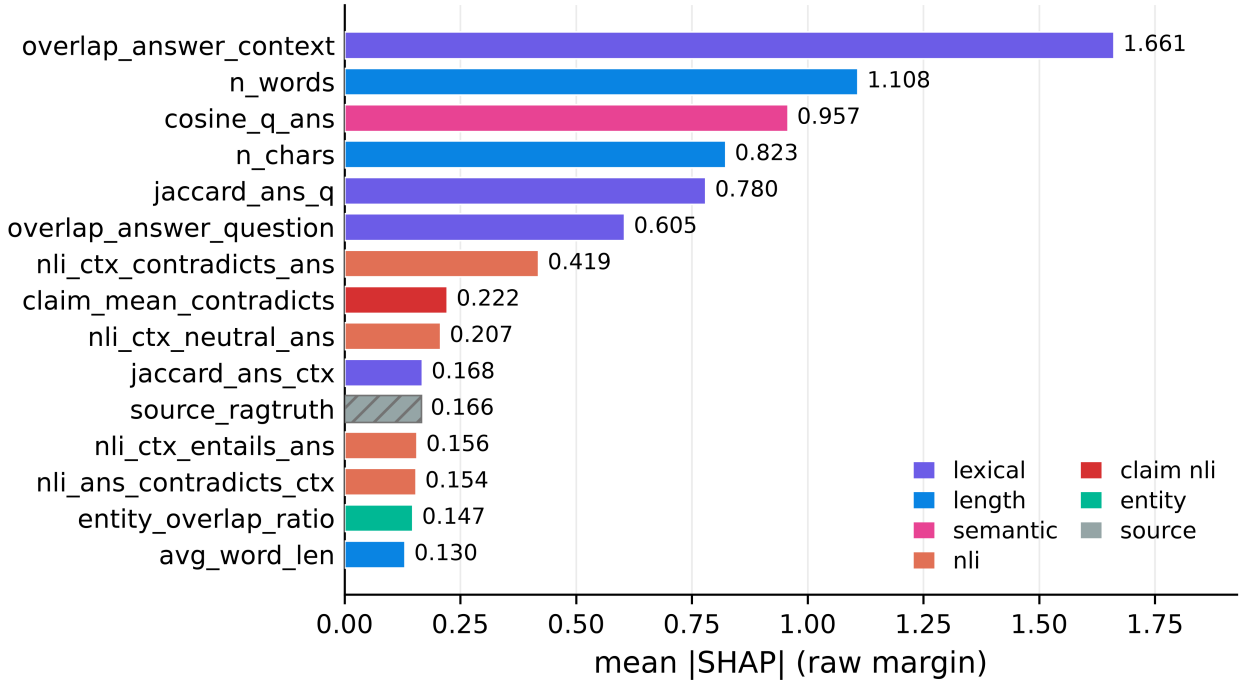


Figure 10: Mean absolute TreeSHAP for the fixed-hyperparameter EC-XGB m3 model, top-15 features (seed 42, 1,000 test rows). Hatching marks a feature that is constant on the sample; its attribution arises through interactions.

is 43.8 ms. The artifact is 0.68 MB, and the marginal cost is about $0.001 per 1,000 predictions. The same 200-row protocol on CPU takes 2.00 s per answer on average in a single contended run (p50 0.98 s, p95 7.56 s), so the 127.5 ms figure depends on GPU feature extraction; the classifier and SHAP steps remain negligible. Feature extraction, not the model, is the cost. Table 16 compares cost and latency: classical heads run in under a millisecond, the m3 pipeline is 127.5 ms, and the two judges are 18 to 24 times slower. The heuristic needs no model and costs almost nothing. RoBERTa-base inference on the same 200 rows takes 6.97 ms per row at batch 8 and fp16, so the m3 feature pipeline is about 18 times slower than the encoder alone. The encoder still needs a fine-tuned classification head, and its training cost is not included.

Table 15
Per-component latency in milliseconds (n = 200).

Component	p50	p95	mean
Core lexical	0.10	0.15	0.11
Entity (spaCy NER)	21.13	47.24	25.34
NLI (2 directions)	27.09	68.60	40.09
Semantic (MiniLM)	9.25	15.55	12.56
XGBoost inference	1.26	8.80	2.62
SHAP explanation	1.91	15.52	3.65
Base total	61.77	—	—
Claim NLI (m3)	43.75	83.21	53.11
Total m3	127.51	234.59	145.89

Note. Components are timed in separate instrumented passes over the same samples, so the component means need not sum to the end-to-end median. The base total is the 26-feature pipeline; the m3 total adds the eight claim-level NLI features. Retrieval and judge calls are excluded.

LLM judges are not fair evaluators, and their preferences can shift with answer order [38]. Table 17 compares EC-XGB with two judges on the 900 RAGTruth QA rows at 17.8% prevalence. GPT-5.6 Luna reaches F1 0.565 and

Table 16
Per-baseline cost and latency.

Baseline	Inputs	Size	p50 (ms)	Cost/1K
Heuristic (overlap)	text	—	0.009	≈$0
Logistic regression	26 feats	1.1 KB	0.001	≈$0.001
Random forest	26 feats	8.4 MB	0.331	≈$0.001
NLI-only	6 NLI probs	—	—	≈$0.001
TF-IDF (answer/full)	text	0.58/4.1 MB	0.014/0.110*	≈$0.001
XGBoost (conventional)	26 feats	0.49 MB	0.005	≈$0.001
RoBERTa-base	512 tokens (timing)	125M params	6.97	training only
EC-XGB (m3)	35 feats	0.68 MB	127.51	≈$0.001
GPT 5.6 Luna judge	API	—	2283.07	$0.234
Qwen 3.5 4B judge	4-bit local	4B params	3049.96	$0

Note. Size is the seed-42 artifact on disk. Feature models share the 127.5 ms m3 pipeline end to end; the model-only inference time is 0.008 ms. The NLI-only head was not persisted, so its cost is the 27.1 ms NLI step plus a linear head of the same order as logistic regression. * marks vectorizer-transform-only timings. RoBERTa timing is encoder-only (batch 8, fp16, 512 tokens, RTX 3060) and excludes the classification head and training; the baseline uses 256 tokens, so 6.97 ms is an upper bound on its inference time. Costs are USD per 1,000 predictions.

AUROC 0.910; Qwen 3.5 4B reaches 0.602 and 0.778; EC-XGB reaches 0.310 and 0.558. Both judges outperform EC-XGB on F1 and AUROC, and the target-fitted length-only baseline at AUROC 0.712 also ranks the QA rows better than the full feature model. GPT costs $0.234 per 1,000 calls and takes 2.28 s; Qwen is local and takes 3.05 s. The feature model runs in 127.5 ms and costs about $0.001, but it is not the better QA detector. Under matched multi-source training, the adapted encoder reaches 0.694 on the seen RAGTruth tasks, above EC-XGB's 0.582, so the encoder is the stronger model on the tasks both have seen. The feature model is less degraded on unseen tasks, but both are near chance.

The length-matched panel changes the ranking. On the 158-row exact-length HaluEval set, EC-XGB reaches F1 0.907 and AUROC 0.956, GPT reaches F1 0.865 and AUROC 0.957, and Qwen reaches F1 0.821 and AUROC 0.863. The feature model therefore matches the stronger judge on ranking, and its seed-mean F1 is 0.042 higher. The paired item-bootstrap interval for GPT's majority-vote predictions against the EC-XGB seed mean is −0.052 [−0.127, 0.024], which includes zero; the AUROC interval is +0.008 [−0.034, 0.048]. The natural-prevalence QA row is a domain-shift result, not a general judge ranking. Prompt choice also moves the judge. On 100 HaluEval rows, GPT's F1 ranges from 0.814 to 0.854 and AUROC from 0.910 to 0.954 across the current prompt, a strict evidence-only prompt, and a concise JSON prompt; the strict prompt is best, but the range is about the size of the model gap on this benchmark.

5.6. Error Analysis and Discussion

On the seed-42 fixed-hyperparameter run, m3 makes 43 errors on the HaluEval test set: 8 false positives and 35 false negatives, an error rate of 1.4%. The WP5 audit covers all eight false positives and 12 of the 35 false negatives, chosen to cover the codebook rather than sampled at random. The 20-case draft assigns 13 content errors and seven audit artifacts. These counts describe the reviewed cases, not the 3,000-row test population. A separate 50-case sheet covers the eight threshold false positives, the 17 highest-scoring negatives, and 25 false negatives; the three authors verified 31 artifact cases (22 grounded over-flags, 7 label-noise, 2 short-answer) and 19 content cases across 12 categories. Table 18 shows why length matters: the hallucination rate rises from 2.1% for one-word answers to 90.2% at five to eight words and 99.2% at 17+ words. False positives concentrate in the five-to-eight bin, false negatives in the short bins, so length is a near-perfect label indicator on HaluEval.

Four controls test the length explanation. A logistic regression on the two length variables reaches F1 0.9457 and AUROC 0.9759. A reweighting control equalizes six length bins with 247 examples each; on that set EC-XGB reaches F1 0.9867 and AUROC 0.9987, against 0.9475 and 0.9758 for a length-only baseline, a gain of +0.039 F1 and +0.023 AUROC. A length-matching control pairs one positive and one negative at each length, giving 79 pairs (158 examples) over 1,000 bootstraps. EC-XGB holds AUROC 0.956 while the length-only baseline falls to 0.521, a point gap of 0.435 (bootstrap mean 0.436 [0.351, 0.522]); the F1 point gap is 0.431 (bootstrap mean 0.440 [0.334, 0.552]). The length cue explains the zero-shot failure but not the in-domain ranking.

Table 17
Judge stress tests: two judges, EC-XGB, length-only, and all-positive references on the 900 RAGTruth QA test rows at natural prevalence (17.8%), and on the 158-row length-matched HaluEval set (50% positive).

Model	P	R	F1	AUROC	Flagged	p50 (ms)	Cost/1K
GPT 5.6 Luna judge	0.404	0.942	0.565	0.910	41.4%	2283	$0.234
Qwen 3.5 4B judge	0.570	0.638	0.602	0.778	19.9%	3050	$0
EC-XGB (m3)	0.183	0.998	0.310	0.558	96.8%	127.5	≈$0.001
Length-only	0.439	0.181	0.257	0.712	7.3%	—	≈$0
All-positive	0.178	1.000	0.302	—	100.0%	—	≈$0
Length-matched HaluEval, 158 rows, 50% positive							
GPT 5.6 Luna judge	0.945	0.798	0.865	0.957	42.2%	2283	$0.234
Qwen 3.5 4B judge	1.000	0.696	0.821	0.863	34.8%	3050	$0
EC-XGB (m3)	0.954	0.865	0.907	0.956	45.4%	127.5	≈$0.001
Length-only	0.515	0.443	0.476	0.521	43.0%	—	≈$0

Note. The upper panel is the 900 RAGTruth QA test rows at 17.8% prevalence. GPT uses default decoding and seeds 42/123/456, mean over three repeats; 2,699 of 2,700 calls parsed, and one empty response was scored as grounded at 0.5. Qwen uses local 4-bit decoding with one seed; it is deterministic at temperature 0. EC-XGB is the seed-mean fixed-hp m3 configuration at the 0.5 threshold; the deployed tuned checkpoint gives AUROC 0.549 and a 97.4% flag rate. The lower panel is the 158-row exact-length matched HaluEval set (79 pairs) from the length control; GPT is the mean over three seeds, Qwen is one deterministic seed, and EC-XGB is the three-seed mean. The lower panel controls length but not domain, so it is a stress test, not a second natural-prevalence estimate. Cost/1K is USD at the recorded token counts and an assumed $1.20/M output price.

Table 18
HaluEval test errors of the conventional B2 model by answer length (counts seed 42, rates three-seed means).

Words	Rows	Hallucinated rate	FP rate	FN rate	Mean score
1	524	0.021	0.004	0.182	0.024
2	514	0.033	0.000	0.176	0.035
3–4	474	0.127	0.002	0.178	0.126
5–8	641	0.902	0.106	0.031	0.888
9–16	600	0.982	0.000	0.003	0.980
17+	247	0.992	0.333	0.000	0.993

Note. Counts are seed 42. Rates and mean scores are three-seed means. The 17+ bin contains only two negative rows. The 5–8 word bin contains 66.7% of the false positives and 18 of the 36 false negatives.

Figure 11 shows the same pattern. The shape explains the zero-shot result: complete sentences are long, so the conventional model reads them as hallucinations. Outside the benchmark, EC-XGB ranks the seen RAGTruth tasks better and the display calibrator gives readable target scores. The ablation leaves lexical and length as the only groups that matter, so the in-domain model is dominantly a length and overlap detector; that is why the conventional model flags 99.9% of natural long answers and EC-XGB still flags 61.3%, calibration cannot repair the ranking, and held-out QA stays near 0.55. The external comparison is limited by protocol: no published HaluEval QA result uses the grouped split, a supervised 13B agent reports F1 0.724 on 100 rows [8], and the benchmark's zero-shot judges report 62.6% to 69.8% accuracy [32]. Those settings differ in supervision, evidence access, and test rows, so they are context rather than a head-to-head bar. The benchmark's generation instruction asks for hallucinated answers about five words longer; the observed gap is 8.5 words, and a length-only logistic regression reaches AUROC 0.976 (Table 5). Any comparison that ignores this construction artifact overstates semantic detection.

Five limits bound these claims. Web-search retrieval quality and end-to-end verdict accuracy are not evaluated; the retrieval stress test uses the known source passage. The judge comparison uses 900 RAGTruth QA rows at natural prevalence and the 158-row length-matched set, which controls length but not domain. The audit is error-enriched. HaluEval is saturated, so its in-domain differences are not resolvable. The deployed display calibrator was selected in

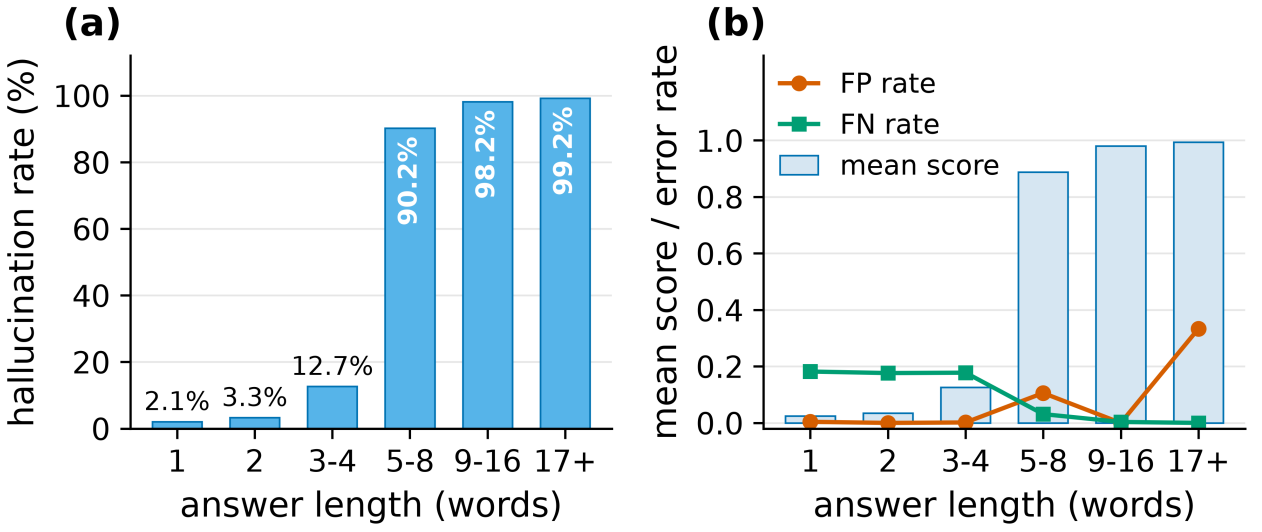


Figure 11: Hallucination rate, mean score, and error rates by answer length for the B2 conventional model.

an earlier fit that compared Platt and isotonic on the test split; the calibration analysis in Table 8 refits Platt without test-based selection.

6. Conclusion

This study tested whether a simple feature model can estimate hallucination risk and how its calibration and ranking change under domain and task shift. EC-XGB is an evidence-consistent XGBoost risk predictor with 26 base features, eight claim-level NLI aggregates, and a source indicator, trained on multi-source rows and paired with calibration and SHAP explanations. On HaluEval QA the fixed-hp model reaches F1 0.9854 and AUROC 0.9986, but lexical overlap and length carry that score: removing them costs 0.0171 and 0.0066 F1, and removing the claim block costs 0.0010. The in-domain score is therefore mostly a length and overlap effect. On RAGTruth the tuned model cuts the pooled flag rate from 99.9% to 61.3%, but the fixed-hp model still flags 96.8% of held-out QA at AUROC 0.558. Target Platt calibration lowers ECE from 0.731 to 0.125, but flags no target rows at 0.5 and loses to a constant predictor on Brier score. Split conformal covers 94.4% at 90%, but positive coverage is only 0.685. On MedHallu the model ranks the hallucinated answer above the grounded one in 86.7% of 1,000 pairs (AUROC 0.821), but flags every row and is badly calibrated (ECE 0.493). The ranking survives the domain change, but the probability does not. Dialogue collapses to an all-positive flag rate, and summarization ranks below a length-only model. One answer takes 127.5 ms on GPU and costs about $0.001 per 1,000 predictions. EC-XGB is therefore a domain-specific risk score, not a general-purpose detector. Future work should test adaptation to unseen QA domains, extend multicalibration over length and claim-count slices, and select thresholds under prevalence shift.

CRediT authorship contribution statement

Md. Atikur Rahaman: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Saiful Alam Sabbir:** Writing - review & editing. **MD. Miraz Ahamed:** Writing - review & editing. **Ohidujjaman Tuhin:** Supervision, Validation, Writing - review & editing.

Declarations

- **Funding:** This research received no external funding.
- **Conflicts of Interest:** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

- **Data availability:** Full raw datasets are not redistributed. The repository includes download scripts, source revisions, SHA-256 hashes, license notes for HaluEval (MIT), RAGTruth (MIT), FaithBench (CC BY-NC-SA 4.0), MedHallu (MIT), and the HaluBench subset (CC BY-NC 2.0), and a 26-case audit file with benchmark excerpts. Splits and feature matrices can be reproduced from the original dataset sources. Model artifacts and the experiment manifest are available from the corresponding author on reasonable request.
- **Code availability:** The training pipeline, FastAPI inference service, and Next.js dashboard are available at `https://github.com/Atik203/HaluRISC` (branch `rebuild`).
- **Ethical approval:** No human-subject data were collected. The three authors annotated public benchmark rows for internal verification; this does not constitute human-subjects research, so ethics approval was not required.

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References

[1] P. Manakul, A. Liusie, M. J. Gales, Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models, in: Proceedings of EMNLP, 2023, pp. 9004–9017. doi:10.18653/v1/2023.emnlp-main.557.

[2] M. Belyi, R. Friel, S. Shao, A. Sanyal, Luna: A lightweight evaluation model to catch language model hallucinations with high accuracy and low cost, in: Proceedings of COLING (Industry Track), 2025, pp. 398–409.
URL https://aclanthology.org/2025.coling-industry.34/

[3] S. Farquhar, J. Kossen, L. Kuhn, Y. Gal, Detecting hallucinations in large language models using semantic entropy, Nature 630 (8017) (2024) 625–630. doi:10.1038/s41586-024-07421-0.

[4] S. Yadav, N. K. Verma, A hybrid framework for hallucination detection in large language models, IEEE Transactions on Artificial Intelligence (2026). doi:10.1109/TAI.2026.3653354.

[5] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, P. Fung, Survey of hallucination in natural language generation, ACM Computing Surveys 55 (12) (2023) 1–38. doi:10.1145/3571730.

[6] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, T. Liu, A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions, ACM Transactions on Information Systems 43 (2) (2025) 1–55. doi:10.1145/3703155.

[7] A. Alansari, H. Luqman, Large language models hallucination: A comprehensive survey, Computer Science Review 61 (2026) 100970. doi:10.1016/j.cosrev.2026.100970.

[8] X. Cheng, J. Li, W. X. Zhao, H. Zhang, F. Zhang, F. Zhang, K. Gai, J.-R. Wen, Small agent can also rock! empowering small language models as hallucination detector, in: Proceedings of EMNLP, 2024, pp. 14600–14615. doi:10.18653/v1/2024.emnlp-main.809.

[9] S. Valentin, J. Fu, G. Detommaso, S. Xu, G. Zappella, B. Wang, Cost-effective hallucination detection for llms, arXiv preprint arXiv:2407.21424 (2024).

[10] D. Sundaragiri, L. M. Reddy, P. Gunavardhan, B. Navahith, Framework for hallucination detection in large language models, International Journal of Engineering Research & Technology (IJERT) 15 (04) (2026). doi:10.5281/zenodo.20025987.

[11] I. Haq, M. Saqib, Y. Zhang, I. A. Khan, Quantifying factual divergence in generative models: Shap-lime based hallucination score for llms, Multimedia Systems 32 (2) (2026). doi:10.1007/s00530-025-02150-4.

[12] Y. Deng, Z. Fang, S. Li, L. Chen, Beyond in-domain detection: Spikescore for cross-domain hallucination detection, in: Proceedings of ICLR, 2026.

[13] F. S. Bao, M. Li, R. Qu, et al., Faithbench: A diverse hallucination benchmark for summarization by modern llms, in: Proceedings of NAACL (Short Papers), 2025, pp. 448–461. doi:10.18653/v1/2025.naacl-short.38.

[14] S. Pandit, J. Xu, J. Hong, Z. Wang, T. Chen, K. Xu, Y. Ding, MedHallu: A comprehensive benchmark for detecting medical hallucinations in large language models (2025). arXiv:2502.14302, doi:10.48550/arXiv.2502.14302.

[15] S. S. Ravi, B. Mielczarek, A. Kannappan, D. Kiela, R. Qian, Lynx: An open source hallucination evaluation model (2024). arXiv:2407.08488, doi:10.48550/arXiv.2407.08488.

[16] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: Proceedings of ICML, 2017.
URL http://proceedings.mlr.press/v70/guo17a.html

[17] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems (NeurIPS), 2017.
URL https://papers.nips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[18] T. Chen, C. Guestrin, Xgboost: A scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD, 2016. doi:10.1145/2939672.2939785.

[19] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, J. Snoek, Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift, in: Advances in Neural Information Processing Systems (NeurIPS), 2019.

[20] P. He, J. Gao, W. Chen, DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing, in: Proceedings of the International Conference on Learning Representations (ICLR), 2023.

[21] S. R. Bowman, G. Angeli, C. Potts, C. D. Manning, A large annotated corpus for learning natural language inference, in: Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2015, pp. 632–642. doi:10.18653/v1/D15-1075.

[22] A. Williams, N. Nangia, S. R. Bowman, A broad-coverage challenge corpus for sentence understanding through inference, in: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), 2018, pp. 1112–1122. doi:10.18653/v1/N18-1101.

[23] N. Reimers, I. Gurevych, Sentence-bert: Sentence embeddings using siamese bert-networks, in: Proceedings of EMNLP-IJCNLP, 2019, pp. 3982–3992. doi:10.18653/v1/D19-1410.

[24] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, M. Zhou, MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers, in: Advances in Neural Information Processing Systems (NeurIPS), 2020.

[25] M. Honnibal, I. Montani, S. Van Landeghem, A. Boyd, spaCy: Industrial-strength natural language processing in Python (2020). doi:10.5281/zenodo.1212303.

[26] D. Ru, L. Qiu, X. Hu, T. Zhang, P. Shi, S. Chang, J. Cheng, C. Wang, S. Sun, H. Li, Z. Zhang, B. Wang, J. Jiang, T. He, Z. Wang, P. Liu, Y. Zhang, Z. Zhang, RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 37, 2024, pp. 21999–22027. doi:10.52202/079017-0692.

[27] J. C. Platt, Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods, in: A. J. Smola, P. Bartlett, B. Schölkopf, D. Schuurmans (Eds.), Advances in Large Margin Classifiers, MIT Press, Cambridge, MA, 1999, pp. 61–74.

[28] B. Zadrozny, C. Elkan, Transforming classifier scores into accurate multiclass probability estimates, in: Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2002, pp. 694–699. doi:10.1145/775047.775151.

[29] A. Gasparin, G. Detommaso, Distance-aware calibration for pre-trained language models, in: Findings of the Association for Computational Linguistics: EMNLP 2024, 2024, pp. 12434–12447. doi:10.18653/v1/2024.findings-emnlp.725.

[30] G. Kelodjou, L. Rozé, V. Masson, L. Galárraga, R. Gaudel, M. Tchuente, A. Termier, Shaping up SHAP: Enhancing stability through layer-wise neighbor selection, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 38, 2024, pp. 13094–13103. doi:10.1609/aaai.v38i12.29208.

[31] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, B. Kim, Sanity checks for saliency maps, in: Advances in Neural Information Processing Systems (NeurIPS), 2018.

[32] J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie, J.-R. Wen, Halueval: A large-scale hallucination evaluation benchmark for large language models, in: Proceedings of EMNLP, 2023. doi:10.18653/v1/2023.emnlp-main.397.

[33] Y. Bang, Z. Ji, A. Schelten, A. Hartshorn, T. Fowler, C. Zhang, N. Cancedda, P. Fung, HalluLens: LLM hallucination benchmark, in: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Vienna, Austria, 2025, pp. 24128–24156. doi:10.18653/v1/2025.acl-long.1176.

[34] C. Niu, Y. Wu, J. Zhu, S. Xu, K. Shum, R. Zhong, J. Song, T. Zhang, Ragtruth: A hallucination corpus for developing trustworthy retrieval-augmented language models, in: Proceedings of ACL, 2024, pp. 10862–10878. doi:10.18653/v1/2024.acl-long.585.

[35] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, et al., RoBERTa: A robustly optimized BERT pretraining approach, arXiv preprint arXiv:1907.11692 (2019). doi:10.48550/arXiv.1907.11692.

[36] C. Chen, K. Liu, Z. Chen, Y. Gu, Y. Wu, M. Tao, Z. Fu, J. Ye, INSIDE: LLMs' internal states retain the power of hallucination detection, in: International Conference on Learning Representations (ICLR), 2024.

[37] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, I. Stoica, Judging LLM-as-a-judge with MT-Bench and Chatbot Arena, in: Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2023.

[38] P. Wang, L. Li, L. Chen, Z. Cai, D. Zhu, B. Lin, Y. Cao, L. Kong, Q. Liu, T. Liu, Z. Sui, Large language models are not fair evaluators, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2024, pp. 9440–9450. doi:10.18653/v1/2024.acl-long.511.